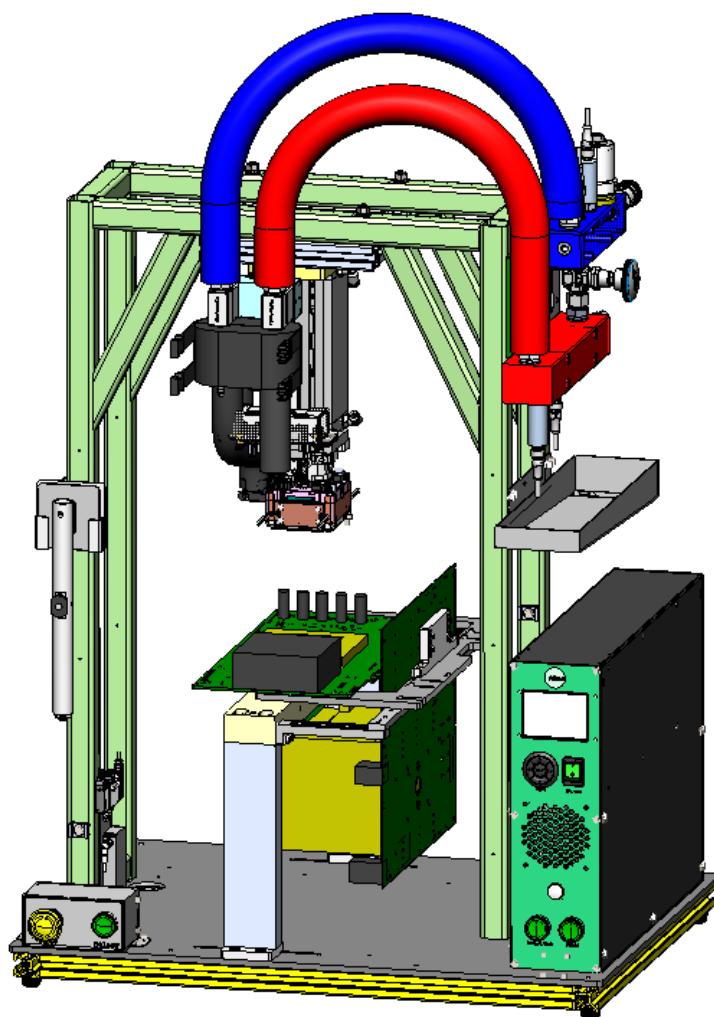




AEM Thermal Station

Operation Manual

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DISCLAIMERS

By using this manual and the equipment, you acknowledge that you have read, understood and agreed to the following:

1. The installation and use of the equipment involves certain risks, and by installing and using the equipment, you voluntarily assume all risks associated with such installation and use.
2. To the fullest extent permitted by law, AEM shall not be responsible or liable for any injury, death, loss, damage, claim, liability, award, judgment, penalty, interest, cost, or expense in connection with or arising from the installation and use of the equipment in non-compliance with this manual, including the misuse of equipment:
 - a) by individuals who are not properly trained in accordance with this manual to operate the equipment,
 - b) that has not been properly set-up or installed in accordance with the specifications/configurations listed in this manual,
 - c) that has not been properly maintained in accordance with the requirements and at the frequency specified in this manual,
 - d) without properly observing the safety precautions, guidelines and instructions listed in this manual, and/or
 - e) which has been modified from the original equipment specifications (whether by AEM's customer or otherwise), including where the equipment contains parts or components not compliant with the original equipment specifications.
3. It is your responsibility to ensure that the use of this manual, including the installation and use of the equipment, is carried out in compliance with all applicable laws, regulations and industry standards, and the terms and conditions of this manual.
4. You agree to indemnify, defend and hold harmless AEM for any loss, damage, claim, liability, award, judgment, penalty, interest, cost, or expense arising out of installation and use of the equipment in non-compliance with this manual, your breach of any applicable law, regulation or industry standard, and/or your breach of any term and condition of this manual, negligence, misconduct, or fraud.
5. AEM reserves the right to modify, update, or discontinue this manual at any time without prior notice. It is your responsibility to review the most current version of this manual (which is available upon request) before using the information contained in this manual.
6. If you have any doubt on the installation and use of the equipment, or any information contained in this manual, please contact your local AEM representative for the site for further clarifications prior to installing and using the equipment or using the information in this manual.

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Revision Record

Manual Version	Issue Date	Description
0	02 Dec 2022	Initial Release

Abbreviations & Acronyms

Term	Definition
ATC	Active Thermal Controller
DE	Die Engage
DUT	Device under Test
DTF	DUT Temperature Following
EGW	Ethylene Glycol Water
GUI	Graphical User Interface
HTF	Heater Temperature Following
LOTO	Lock Out/ Tag Out
MSD	Musculoskeletal Disorders
PD	DUT Power
PID	A transfer function consists of Proportional, Integral, Differentiator
PPE	Personal Protective Equipment
PSET	Set Power
PWM	Pulse Width Modulation
QD	Quick Disconnect



Term	Definition
RTD	Resistance Temperature Detector
SSR	Solid State Relay
SW	Software
TCO	Thermal Cutoff
TCS	Temperature Control System
TD	Dut Temperature
TSET	Set Temperature
TSP	Temperature Set Point
TTV	Thermal Testing Vehicle
UI	User Interface



NOTES

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1.Introduction



1.1 Characteristic of Equipment

1.1.1 User Friendly Graphical Interface

The AEM Thermal Station is equipped with a touch screen that has a user friendly graphical interface. Through the graphical interface, the user will have full control of the tool and functionality (depending on the “user-level” granted). Testing condition of the tool will be visually displayed on the interface screen to assist the user to navigate through the tool.

1.1.2 Tri-Temperature Testing

The AEM Thermal Station supports SLT testing in tri-temperature test scenarios (cold test, room test, hot test) without compromising the physical integrity of the device under test.

1.1.3 Quick Kit Change Design

The AEM Thermal Station provides better accessibility design to allow user to exchange the kit more efficiently.

1.2 Equipment Configuration

This section shows the AEM Thermal Station overview.

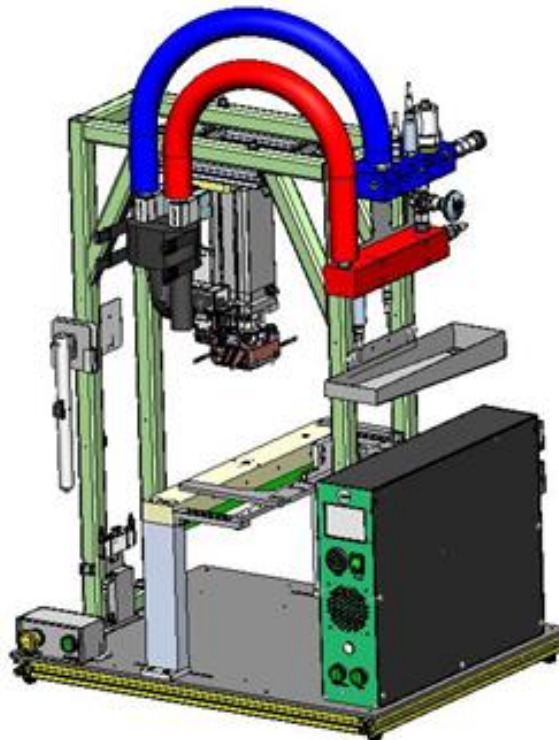


Figure 1-1 AEM Thermal Station front ISO view

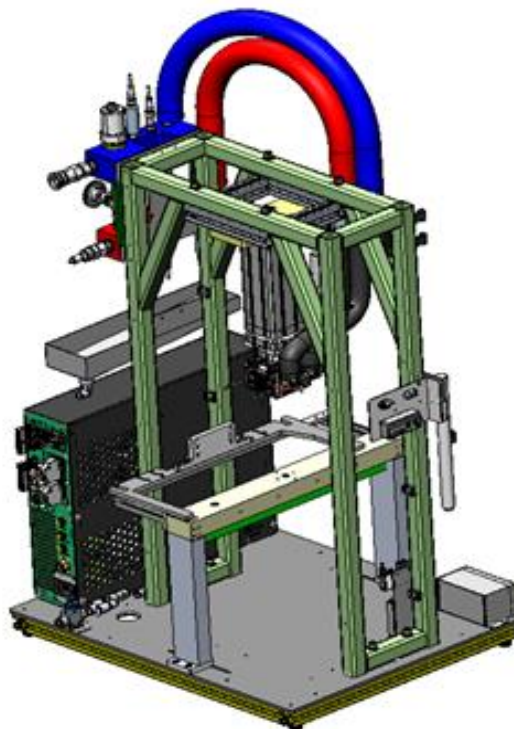


Figure 1-2 AEM Thermal Station back ISO view

1.3 Transport, Handling & Storage

Handling

The equipment will be handled by AEM and AEM authorized personnel from build to packing and crating into shipment package. Shipment and forwarding of the equipment will be handled by customer authorized and appointed forwarder. Customer, customer authorized personnel, and AEM staff onsite or local representative will handle the equipment from uncrating/unpacking, installation, operation, and maintenance.

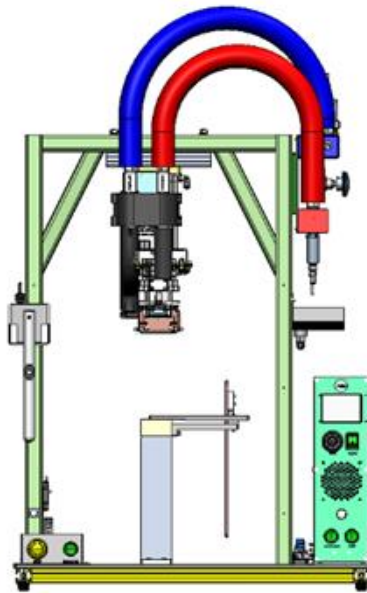


Figure 1-3 AEM Thermal Station

Transportation

Equipment will be transported either via land transport, air transport, or via cargo container ship for sea transportation. The transport mode depends on the customer's site location, delivery time as well as the price. Customer appointed and authorized cargo forwarder will handle the shipment and transportation based on customer's request and direction. During transportation, altitude depends on the mode of transportation and all necessary requirements must be adhered by the forwarder. Equipment should be kept in a controlled environment (temperature and humidity) during transportation. Refer to [Table 3-6 Mechanical specification](#) for relative humidity, machine storage humidity, and machine storage temperature.

Storage

Equipment will be stored at AEM facility until it is picked up by the customer appointed forwarder. Forwarder will store the equipment at their warehouse once picked up from AEM facility until the equipment is shipped out. Customer will store at their own designated warehouse according to their onsite location until



the equipment is uncrated and moved into the production floor. Equipment should be kept in a controlled environment (temperature and humidity) during storage, with altitude not exceeding 1000 m above sea level. Refer to **Table 3-6 Mechanical specification** for relative humidity, machine storage humidity, and machine storage temperature.

General Recommendations

Equipment should not be subjected to vibrations. Ensure tip and fall shock indicators are not triggered due to vibrations or mishandling. If any shock indicators are triggered, contact AEM Singapore representative immediately for further support.



NOTES

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2. Safety Precautions



2.1 Safety

Make sure that every person who handles the equipment (teaching, servicing, maintenance, programming, inspection) are appropriately trained in every aspect. Only service personnel who are by reason of their training; experience as well as their knowledge in the relevant standards; regulations, are qualified and empowered to carry out necessary safety work in the plant.

Re-training of operator and technical personnel shall be required when equipment system calls for any changes.

2.1.1 General Safety Information

The AEM Thermal Station consists of various safety features to prevent the operators and maintenance personnel from injuring themselves or others and damaging the machine. Mechanical and electrical safety hazards may occur for any automated machines. The safety features for the AEM Thermal Station consists of circuit breakers and other devices that allow safe operations and maintenance of the equipment. These features will be further highlighted in the following chapters.

Purpose

To establish safe work practices with potential hazards.

Scope

Safety applies to all workers who perform tasks in a potentially hazardous environment during installation, operations, modifications, maintenance, and repair of machinery and equipment. Operations and maintenance of any automated equipment exposes personnel to potential hazards. Properly following the procedures set forth in this section will ensure safe operations and maintenance of the equipment.

2.2 Safety Alert Boxes

While using this manual, be sure to observe the alert boxes and symbols shown below.

2.2.1 Safety Label Precaution

All hazardous labels are to alert personnel accessing to the area / cabinet to perform work. It does not isolate potential hazards from the personnel as such; service personnel must take the necessary safety procedures (e.g., power off) to ensure the system is fully isolated from the facility's energy supply. Failure to heed precautions can result in injury to people or damage to machine.

Always heed the information provided by any labels!

The following conventions are used to indicate and classify precautions in this manual.

Table 2-1 Safety label description

  	Indicates an imminently hazardous situation, which, if not avoided, will result in death or serious injury. (e.g., moving parts, high voltage electric parts) Danger alert boxes are Red.
 	Indicates a potentially hazardous situation, which, if not avoided, could result in death or serious personal injury. Warning alert boxes are Orange.

 ⚠ WARNING HOT SURFACE INSIDE. Contact may cause burn. Do not touch, or wear protective gear before servicing internal parts.	
 CAUTION  PINCH POINT	Indicates a potentially hazardous situation, which, if not avoided, may result in minor or moderate personal injury and / or damage to the equipment. Caution alert boxes are Yellow.

2.2.2 General Safety Guidelines

The following are some of the hazard(s) that a worker may face while working on the tool.

Table 2-2 Type of hazards

   	Electrical Hazards: Hazardous voltages may exist and can cause electric shock or burns. Possible injury or death can occur. Only qualified personnel should service this equipment. Perform equipment or module power down and power off before servicing. Before beginning work, always evaluate and perform power off on any module with hazardous energies that service personnel will be exposed to for the scope of work. Be aware that hazardous voltages will still be present on the line side of any isolator switch.
  	Mechanical Hazards: These hazards exist in cases where a component can move in such a way as to impact or otherwise injure personnel performing maintenance functions on the equipment. To avoid these hazards, always operate the equipment with the safety covers in place and door interlocks enabled. When performing maintenance functions with the covers open, be sure to avoid those areas where impact with the moving parts of the equipment is possible. When removing components, it may be necessary to isolate potential mechanical energies by restraining the mechanism using cable ties or blocking.

	Compressed Hazards: OFA is supplied to the equipment for many functions. Even though the equipment is electrically powered down, pneumatic pressure is still present. When performing maintenance functions on pneumatic components, always isolate the OFA by closing the manual valve and safely bleed the pressure out of the system.
	Thermal Hazards: Actuation Pan & Bottom Pan can also operate at elevated temperatures and capable of causing burns. Be sure that the components to be worked on are not at elevated temperatures before performing maintenance functions.
	Usage of Material Handling Equipment: When operating or working near Material Handling Equipment (hoists, pallet jacks, forklifts, and overhead cranes) exercise caution to avoid injury. Safety boots (steel toes) are mandatory. Be aware of nearby personnel and equipment and avoid contact. Work with site personnel for proper preparation and reporting prior to employing Material Handling Equipment if required.
	Hand and Portable Power Tools: Exercise caution when using power tools. Safety glasses are mandatory. Hearing protection may be needed depending on the tools used and noise level. If the noise level is above 80dB, earplugs are mandatory. Always power down the equipment and perform before using power tools on the equipment. Use a vacuum system to reduce the spread of particles or debris. Work with site personnel for proper preparation and reporting prior to employing power tools if required.
	Fluid Hazards: When working on EGW lines, isolate the fluids from the section being worked on by closing valves or disconnecting QD's. Drain the section if needed.

	House Keeping: When possible, perform maintenance without blocking aisle ways. If necessary, erect warning signs and / or barricades to warn / block the necessary space to safely perform the required maintenance. Do not block emergency exits. Work with site personnel for proper preparation and reporting prior to work requiring blocking aisles way if required.
	Ergonomics: Avoid Musculoskeletal Disorders (MSD). This range of disorders may be avoided by using the correct tools, proper postures, stretching and avoiding repetitive activities or sitting for prolonged periods. Know your limits and seek assistance when needed. The symptoms of MSD include tingling, numbness, burning sensations, weakness, swelling, soreness, stiffness, spasms, pain, reduced range of motion, aches, and cramping. At the onset of symptoms, notify a supervisor. Evaluation by medical professional may be required.
	PPE: Standard Personal Protective Equipment (PPE) include shoes (closed toe, closed heel, and non-porous), gloves (latex or nitrile), and eye wear (safety, side protection). Other PPE may be required for specific locations or activities.
	Control of Hazardous Energy: Failure to follow safety protocols and reporting requirements are punishable by disciplinary action up to and including termination.



2.2.3 Personal Safety Equipment

Installation or de-commissioning of AEM Thermal Station requires full preparation in advance. Installation team must fully understand the safety precautions for machine handling as well as the factory health and safety regulations.

1. PPE

Standard Personal Protective Equipment is required as listed below:

- Safety goggles (non-conductive frame)
- Safety boots (steel toe)
- Smock / Jump suit
- Hard top helmet / bump cap
- Safety glove
- Rubber / leather / cotton glove

2. Tools and Equipment

Other than basic hand tools, the team needs to prepare the following equipment for machine installation:

- Multi-meter



2.3 Safety Information

The electrical machines are operating appliances for use in current heavy industrial facilities. When being operated, the machines have dangerous electrical live parts, and in some cases moving or rotating mechanical parts as well. They can therefore cause serious injuries or damages.

Make sure personnel does not perform unsafe work practices that include unqualified personnel removing covers, using tools in manners that are inconsistent with this manual, operation of the machine by an untrained individual and removing safety features.

Those responsible for safety in the plant must therefore guarantee:

- Only qualified people work with the machine.
- Machine manual delivered with the machine is always available to such people and that service personnel are obliged to observe the procedures contained therein.
- Unqualified persons are forbidden to work at or near the machine.
- Person working on a tool can locate and identify all safety warning labels before use.
- Operators do not wear loose clothing or jewelry while operating the system.
- Main power is off before opening A/C cabinets.
- Operators of the machine are aware of stop locations and operation in case of system malfunction.
- Earth wires are not disconnected from the machine.
- Internal parts of the controller to be never touched during operation.
- Wires and / or connectors are not unplugged when the power is on.
- Machine is not connected to a voltage or frequency other than what is indicated in the specifications.
- Power to the machine is not cut-off during production.

Qualified people are, by reason of their training, experience, and these instructions, as well as their knowledge of the relevant standards; regulations; accident prevention regulations and operating relations, have been authorized by people responsible for plant safety to carry out the necessary work and are able to recognize and avoid possible dangers.



Note: To prevent breakdowns, it is necessary that the prescribed maintenance, inspection and revision measures be carried out regularly by experienced servicing staff (see above).

Changes to normal operations (increased power consumption, higher temperatures, increased noises etc. or responses from the monitoring equipment) can indicate that the tool functions are impaired. To prevent breakdowns, which can directly or indirectly lead to serious injuries or damages, the maintenance staff must be informed without delay.

Unauthorized modification of the tool can lead to personnel injury and machine damage.

If abnormal conditions are observed, activate the stop on the machine immediately.

If in doubt, switch the machine off immediately and contact equipment owner at site for further support.

For any Environment, Health, and Safety related issue, contact local EHS representative at the site for immediate support.

For any tool and equipment related issues which may need attention of AEM, contact local AEM representative for the site or AEM Singapore at +65 6551 0315.

2.4 Safety Devices

In this machine, all possible sources of accidents have been considered and can be avoided using adapted protection devices.

2.4.1 Hazard Labels & Locations

Safety labels are found on the doors and cover parts to draw attention to hazards present behind or under those protections.

All intervention in these zones must be done only by qualified service personnel, who are trained for this task and are responsible for the maintenance of this machine.

All hazardous labels conform to ANSI Z535 requirement.

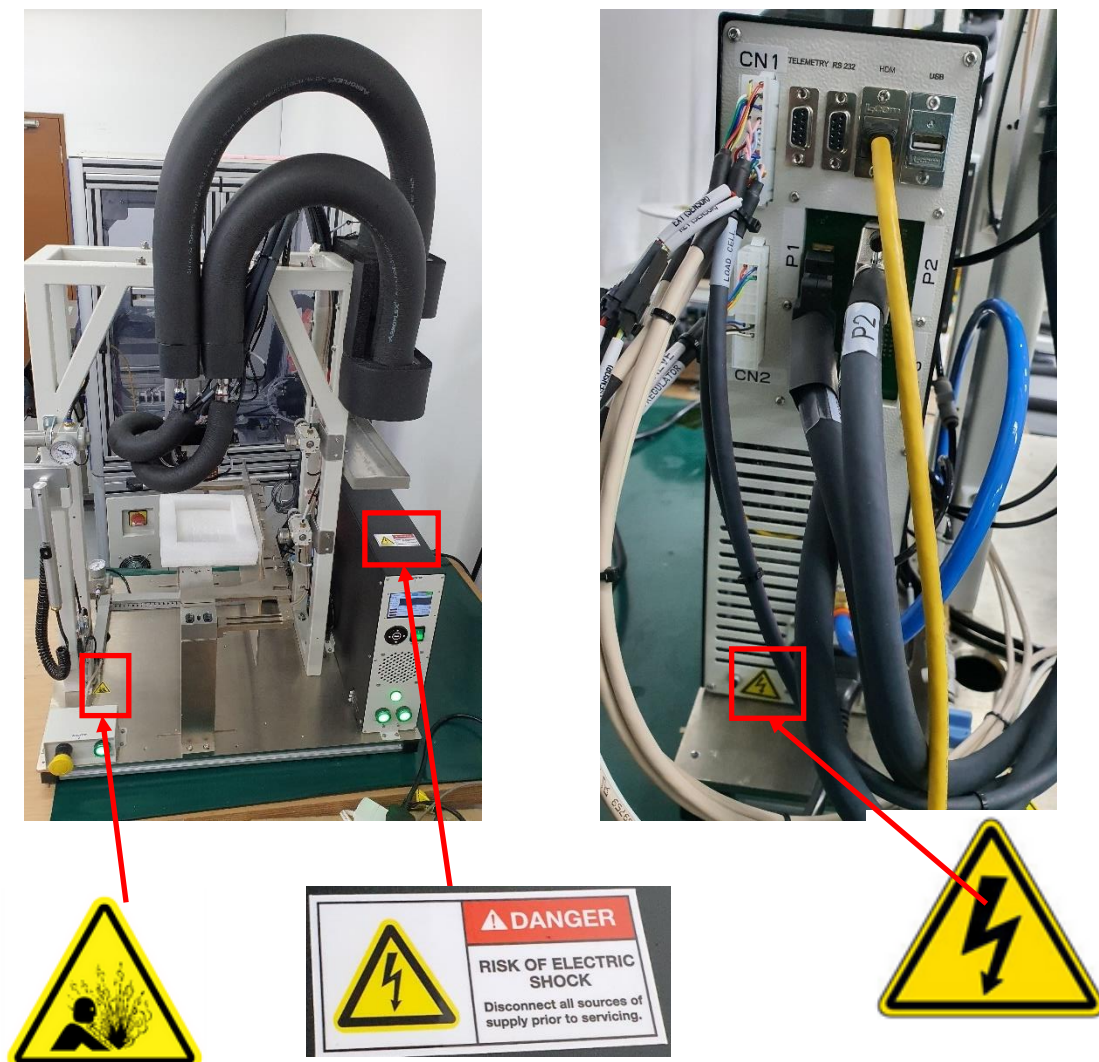


Figure 2-1 Hazard labels & location



2.5 Safety Hazard

Important safety information for handling and operating the equipment such as information about installation, operation, maintenance, servicing, and inspection is provided in each chapter.

2.5.1 Potentially Hazardous Energy Sources

Table 2-3 Potentially hazardous energy sources

Electrical Power Supply					
	Electrical Power Supply	Voltage	Max Current	Phase	Frequency
01	AEM Thermal Station	230VAC	13A (3910W)	Single	50/60Hz
02	Main EGW Supply Flow Rate	4 LPM (minimum), -30 °C temp			
03	Main EGW Supply Pressure	7 Bar			
04	Main Air Supply	5.5 – 9 Bar, oil free, -30C or lower dew point			

2.5.2 Precaution Before Automatic Operation Mode

- Keep all personnel away from the machine as the movement could cause serious injury when struck by moving parts.
- Ensure that the machine is free from any foreign objects.
- Ensure that all panel doors, which are equipped with safety interlock, are closed.



2.5.3 Hazardous Energies Listing

Table 2-4 Type of hazardous energy

YES	NO	Energy Type	YES	NO	Energy Type
		1) Electrical		√	8) Ionizing Radiation
√		a) Low voltage (50 – 600V)		√	9) Non-ionizing
	√	b) High voltage (>600V)		√	a) Ultra-violet
	√	2) Chemical - Explosion, pressure, extreme heat, fire, corrosive, reactive, oxidizer, toxic		√	b) Infra-Red
√		3) Pressure (hydraulic/pneumatic)		√	c) RF / Microwave
	√	4) Vacuum		√	d) Laser
√		5) Mechanical – Capable of crushing, pinching, cutting, snagging, striking		√	e) Magnetic
√		6) Thermal (high temperature) – Surface temperature, hot liquids, steam		√	10) Stored – Flywheels, springs, differences in elevation, elevated parts that could drop, capacitors, batteries
	√	7) Thermal (cryogenic) – Contact with super cold surface or with a cryogenic liquid			

2.5.4 Draining of EGW Line

Personnel must take note that there may be EGW hazard when performing service or maintenance. Personnel must shut off the EGW inlet valve at customer's facilities, cutting off incoming EGW sources before draining EGW hoses. Unplug the EGW inlet/outlet quick connect and reconnect to the QD for air supply and reservoir tank to drain and purge all EGW inside the thermal head.

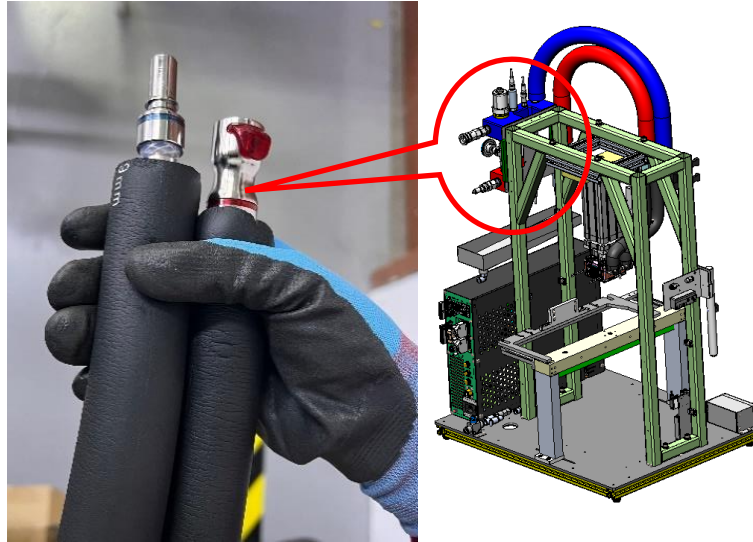


Figure 2-2 EGW draining location behind the station

2.5.5 Main Power High Voltage Supply Hazard

This equipment contains high voltage supply in certain areas of electrical components. This is a hazardous situation where it can cause serious material damage or carry a risk of personal injury.

In consequence, it is strictly forbidden to modify, add or take off any parts and/or other elements that constitutes the machine.

Those responsible for safety in the plant must therefore guarantee:

- That only qualified persons work with the machine.
- That the machine manuals delivered with the machine are always available to such persons and that they are obligated to observe the procedures contained therein.



2.5.6 Standard 3 Point Safety Check

When checking the voltages within the tool, always validate the multimeter being used by performing the following checks:

- Set the multimeter to the function to be used in the tool.
- Set the range for the measurement to be used.

Note: The multimeter should indicate 0 volts (or amperage or infinity resistance).

- Measure a known working voltage source such as a working power outlet.

Note: The meter should indicate normal supply voltage:

- 230 VAC at United States.
 - 230 to 240 VAC at Singapore & Malaysia.
 - 220 VAC at China, Israel & Vietnam.
- Do not continue until the correct voltage is indicated.
 - Switch to another multimeter if needed or try a different outlet.

Note: This verifies that the multimeter is functional before performing the test measurement.

- Perform the test measurement in the tool.
- Record the reading if needed.
- Again, measure a known working voltage source such as a working power outlet.

Note: The meter should indicate normal supply voltage (See above for normal supply voltage).

Note: This verifies that the multimeter is functional after performing the test measurement.



2.6 Environmental Consideration

Important environmental consideration for handling materials used in installation, operation, maintenance, servicing, and inspection is provided in each chapter.

2.6.1 Warning of Using Isopropyl Alcohol

Isopropyl Alcohol (IPA) is a highly flammable liquid & can form an explosive mixture with air at elevated temperatures. It can cause eye & skin irritation if under contact in either vapour or liquid form. Follow the following precautions before & during use of Isopropyl Alcohol.

Note:

- The maximum concentration of IPA is 10%.
- Wear hand gloves (chemical resistant) while using IPA.
- Wear chemical goggles for eye protection while using IPA.
- Pour about 5 to 10 ml of IPA into a disposable container & use cotton bud to clean the parts specified for preventive maintenance requirements.
- After cleaning the parts, dispose of the container & cotton buds by sealing in a plastic cover. Need to ensure the disposal methods are in accordance to the local environmental regulation norms and/or Customer Site Disposal Procedure.
- Ensure IPA is handled in a well-ventilated area.
- For further information on IPA, refer to [12.1.1 Isopropyl Alcohol](#).



NOTES

[illegible]



3. System Overview



3.1 Facility Requirements

3.1.1 Electrical Requirements

Table 3-1 Electrical requirements

Item	Voltage	Wiring Configuration	Full Load (Amps)
AEM Thermal Station	230 VAC	P+N+E, SINGLE PHASE	13A

3.1.2 Gas Requirements

Table 3-2 Gas requirements

Item	Connection Type	Internal Regulator	Operating Pressure (PSI)
	Connect Size	Internal Shutoff	
OFA -2-9BAR	One Touch Fitting	YES	100.0
	10 mm Horse	YES	

3.1.3 Liquid Requirements

Table 3-3 Liquid requirements

Item	Connection Type	Operating Pressure (PSI)	Average Flow (LPM)
	Connect Size		Peak Flow (LPM)
EGW Supply	Quick Connect	70 BAR	4 LPM
	SS-QC8-D-810EP		
EGW Return	Quick Connect	70 BAR	4 LPM
	SS-QC8-B-810EP		

3.1.4 Drain Requirements

Table 3-4 Drain requirements

Item	Connection Type	Connect Size	Internal Shutoff	Average Flow (GPM)
				Peak Flow (GPM)
EGW	Quick Connect	SS-QC8-B-810EP	-	-
				-

3.1.5 Equipment Footprint

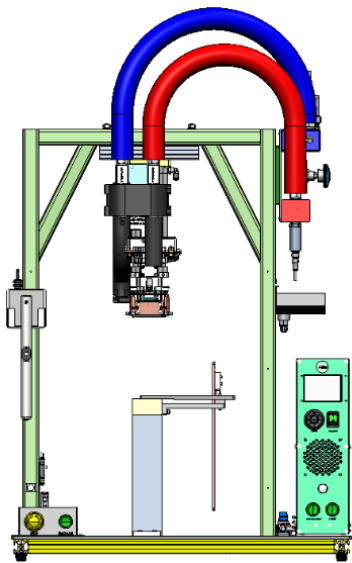


Figure 3-1 Equipment footprint

Table 3-5 Equipment footprint

Dimension	Value
Length (mm)	700 mm
Width / Depth (mm)	540 mm
Height (mm)	1043 mm

3.1.6 Shipping Containers

Containers Dimensions:

Dimension	Value
Length (mm)	900 mm
Width / Depth (mm)	790 mm
Height (mm)	1380 mm
Weight (kg)	81 kg



Figure 3-2 AEM Thermal Station container



Figure 3-3 Loose part packaging

3.2 Mechanical Specification

This section provides the mechanical specification required to operate the system.

The table below lists the mechanical specifications. These specifications set a standard for operations. Equipment performance is gauged by the operation specifications set here.

Table 3-6 Mechanical specification

S/NO	ITEM	SPECIFICATION
1	Operating Temperature	12°C to 18°C
2	Relative Humidity	40% - 60%
3	Maximum Noise Level	65 dB
4	Machine Colour	White + Black
5	Machine Weight	62KG
6	Altitude	Up to 1000 m
7	Machine Storage Temperature	-25°C to 55°C
8	Machine Storage Humidity	5% - 95%. Atmosphere must always be non-condensing

3.3 Equipment Size

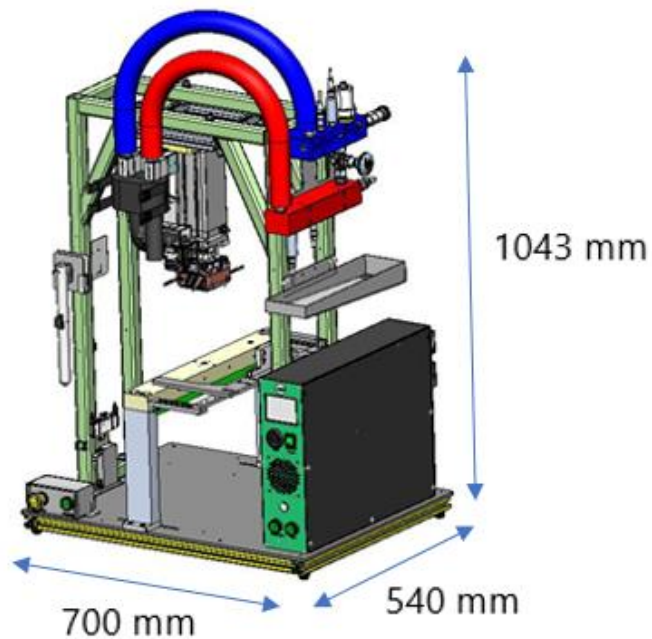


Figure 3-4 Dimension of AEM Thermal Station

3.4 AEM Thermal Station Component

The AEM Thermal Station houses the Thermal Head, Control Box and Test Board to perform the System Level Test.

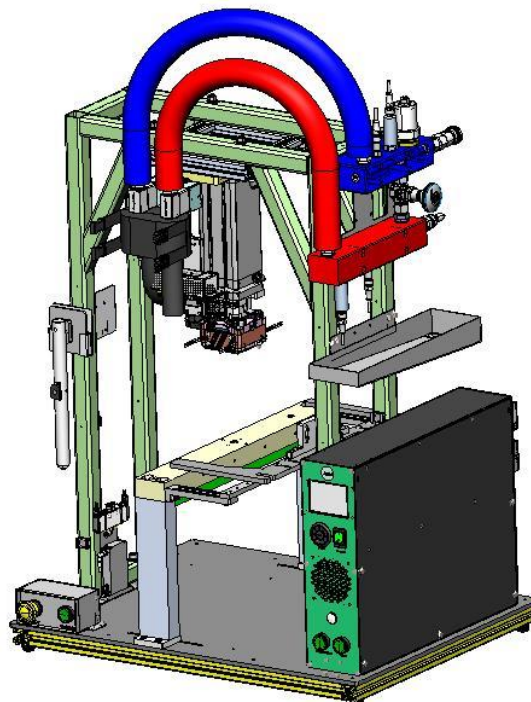


Figure 3-5 AEM Thermal Station front view

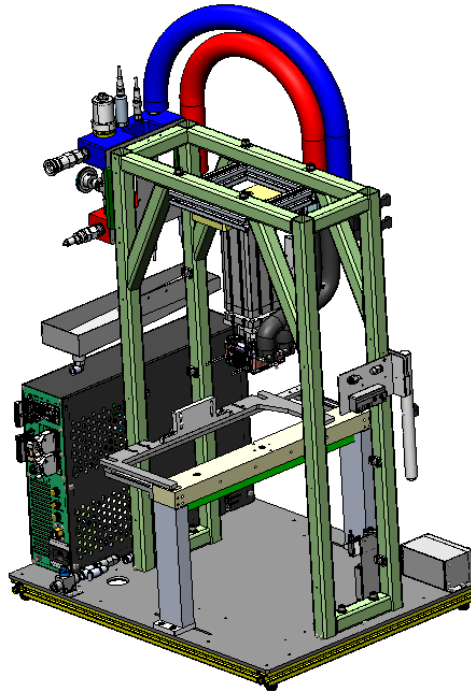


Figure 3-6 AEM Thermal Station rear view

The major components of AEM Thermal Station and their locations are as shown below.

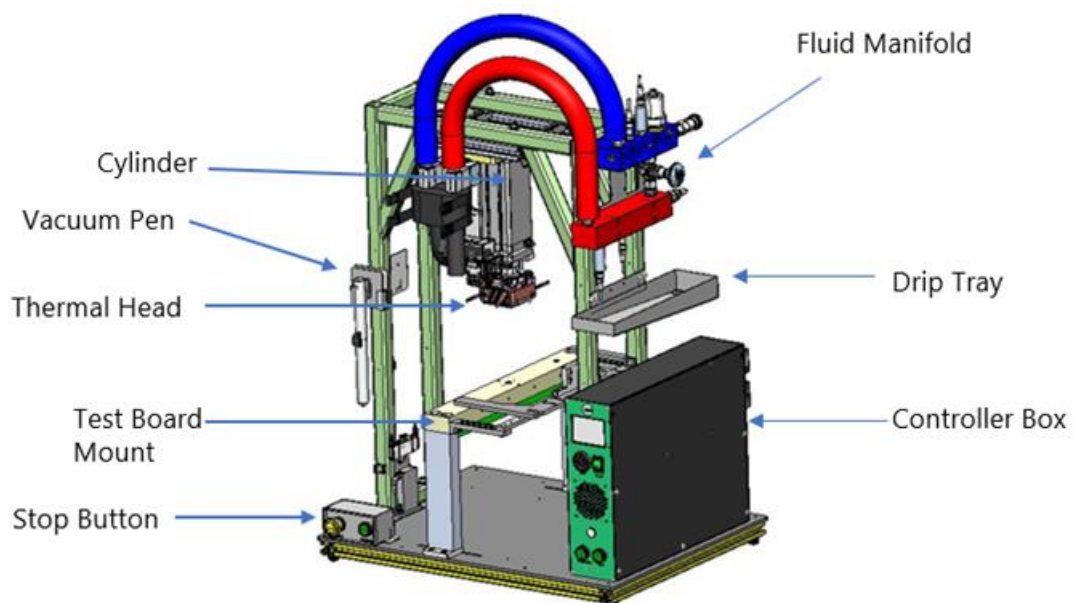


Figure 3-7 Components of AEM Thermal Station



Table 3-7 Components of AEM Thermal Station

Component	Description
Air Purge Circuit	Compressed air to purge EGW from system
Thermal Head	Liquid Cooled heatsink 1 zone heatlet (total 1600W heater power) 1RTD feedback Dual force- die level & substrate level
Water Supply & Return	Supports supply of -30C Insulated manifold and hose
Thermal Controller	Micro-controller FPGA thermal engine 3000W power supply Supports 8 heatlet channels Touchscreen with navigation button
Test Board + Socket Assembly	Contains the test motherboard and TTV that holds the DUT

3.4.1 Controller Box

The Controller Box provides thermal and electronic pressure control.

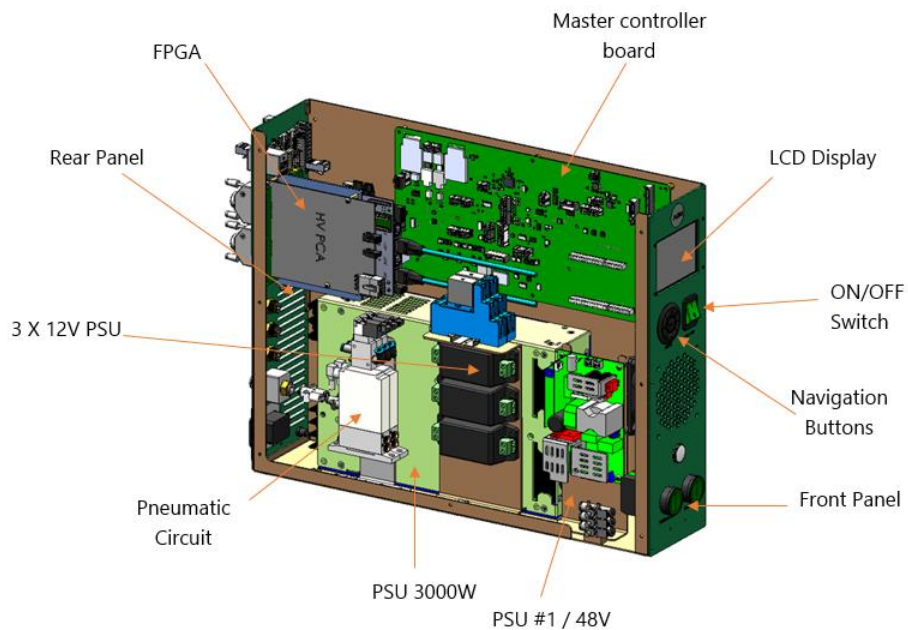


Figure 3-8 Controller box (without cover) components

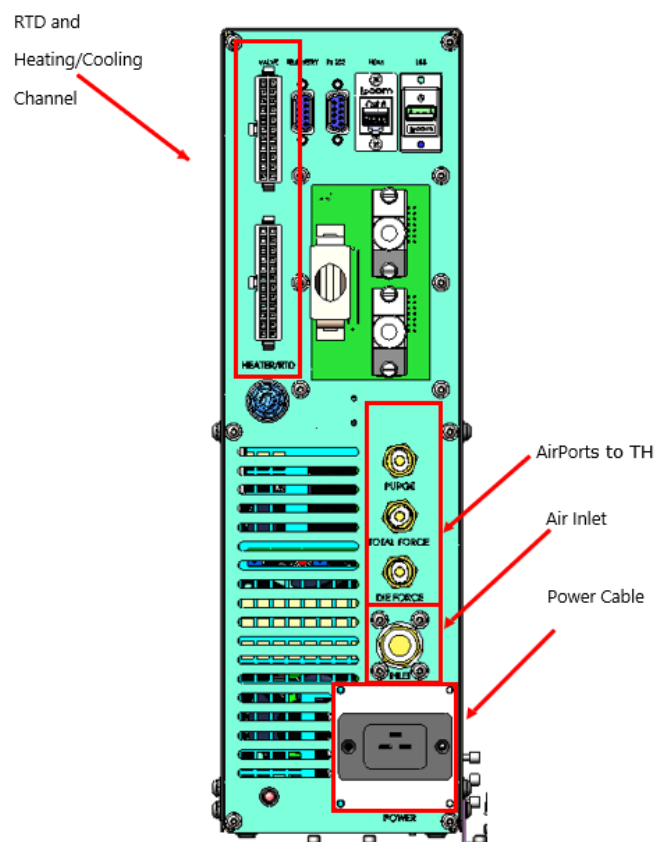


Figure 3-9 Controller box rear panel

3.4.2 Thermal Head & Kit

The kit includes the nest base with top and bottom layers, the heatsink, heat let and the PCB connectors on the kit.

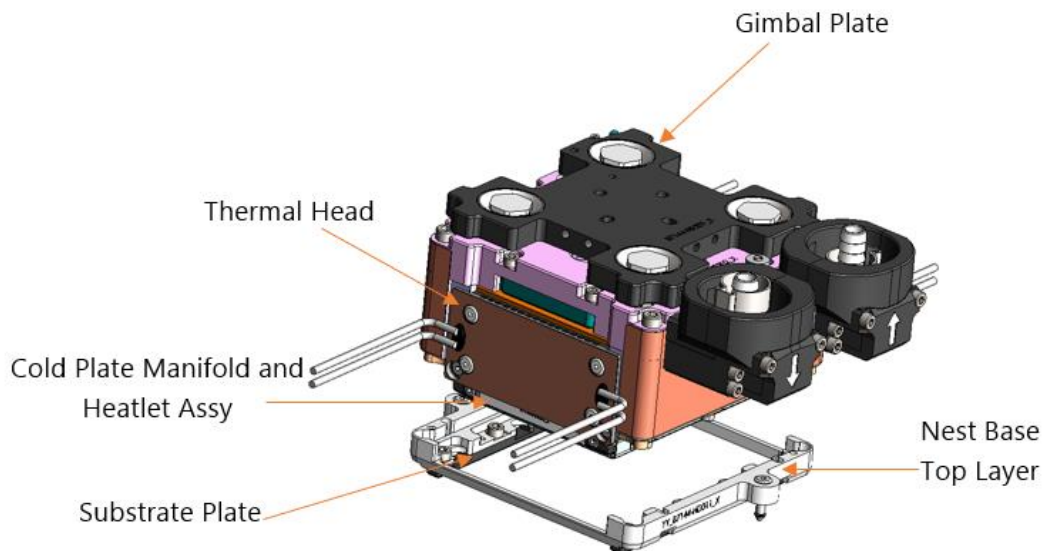


Figure 3-10 Thermal Head Components

3.4.3 Heatlet with RTD

PWM function for heater and monitors temperature (-25°C to 125°C) with the RTD within $\pm 0.5^{\circ}\text{C}$ tolerance. The heatlet has a safety feature that automatically cuts off heater power if temperature exceeds 144°C .

Note: There is a 1wire –EEPROM to save the information of the kit ID at the connector of each kit.

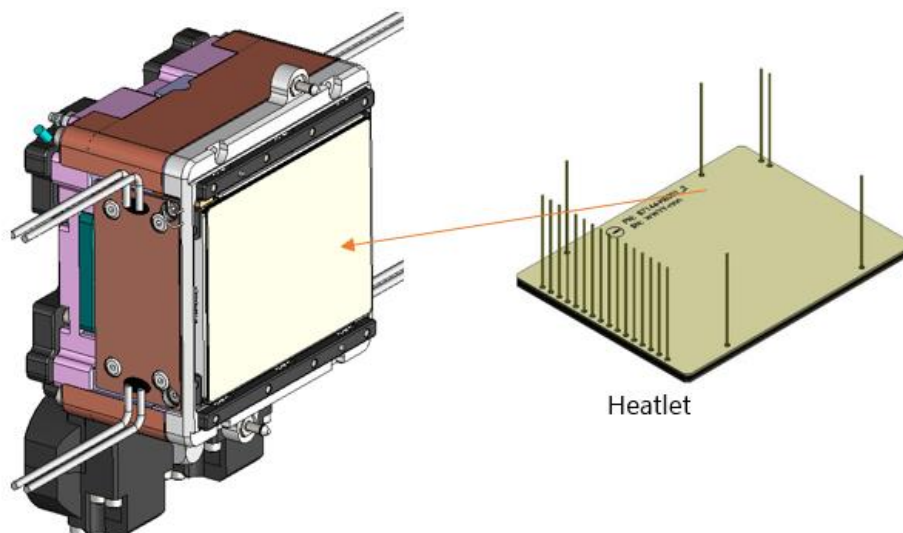


Figure 3-11 Thermal Head with Heatlet

The heatlet single zone with a total of $200 \times 8 = 1600W$. 1 RTD is used to measure the temperature.

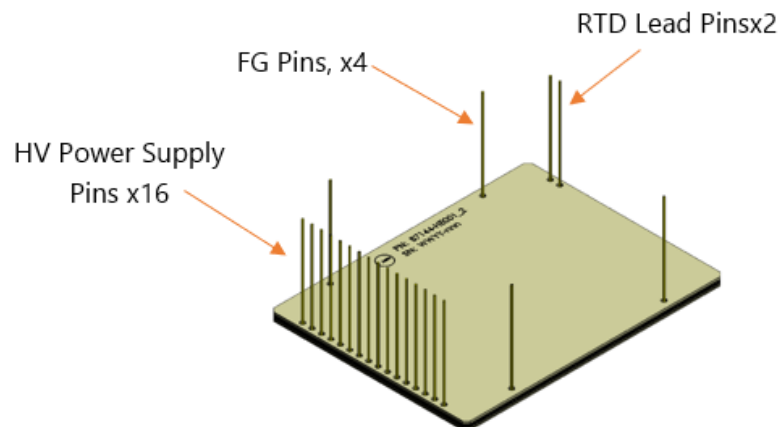


Figure 3-12 Heatlet

3.4.4 Air Purge Push Button

Air purge push button to set purging to auto mode or manual mode. Comes with orifice adjustable flow control (air purge and cylinder stroke).

Air Purge Auto / Manual Push Button

a) Auto Mode

Purge turns on when Hose Heater senses thermal head temperature higher than the set value (On Temperature). System will continue purging until Temp reaches user assigned value (Off Temperature), after which it will turn off.

b) Manual Mode (When button is depressed)

Purge is always on when in auto mode and manual mode.

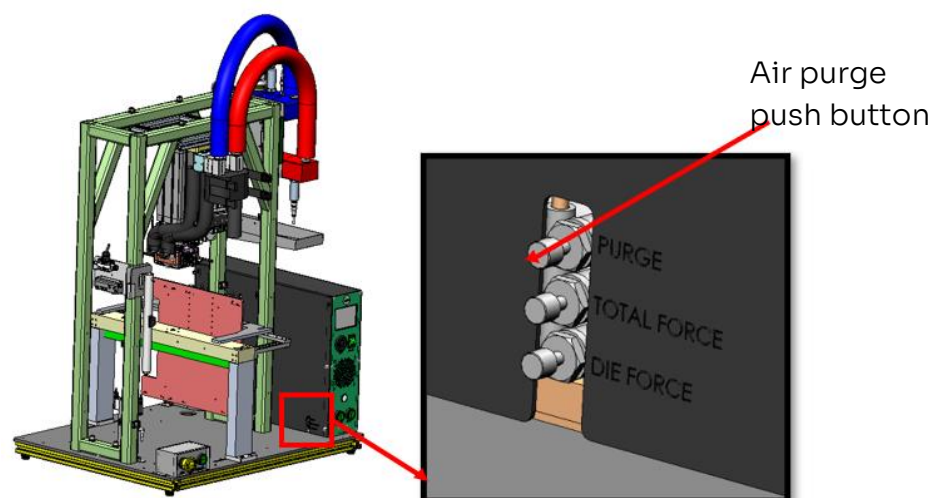


Figure 3-13 Air purge push button



NOTES

[illegible]



4. Installation

4.1 Installation

1. Check the condition of AEM Thermal Station for any major damages and ensure all the components are in good condition.



Figure 4-1 AEM Thermal Station

2. Connection layout: Check facility supply and connect specified quick connect to facility.
3. Make sure the connections of rear panel are correct.

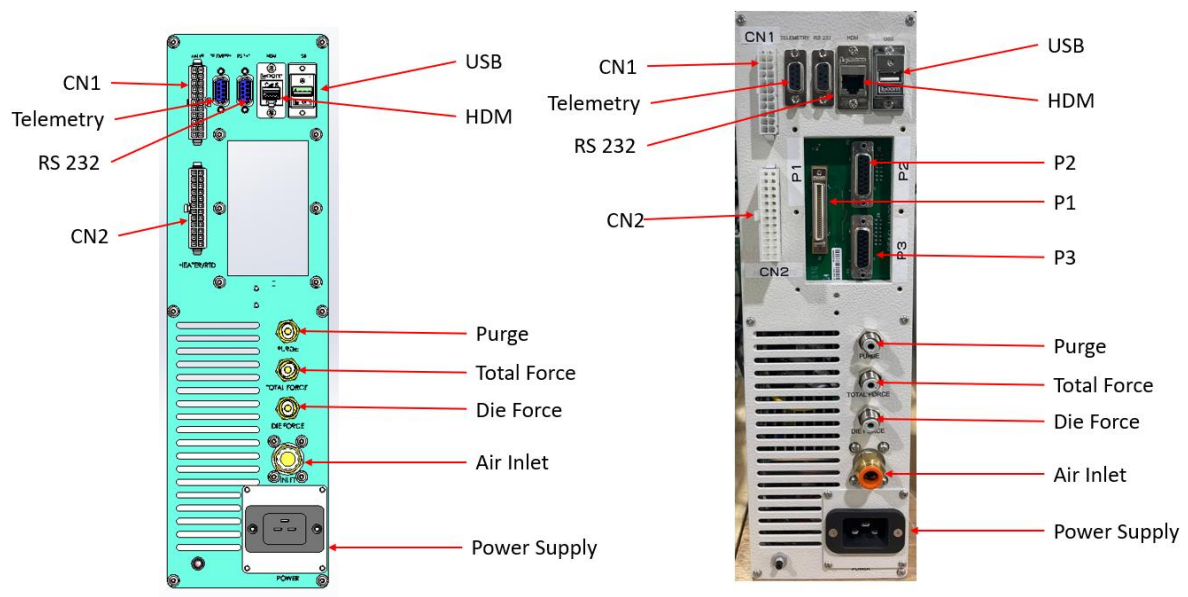


Figure 4-2 Connection and example after connection



NOTES

[illegible]



5. User Interface

5.1 Network Topology

Note: If GUI is required, LAN Cable must stay connected with a powered-ON PC/Mac/Router before power on multiple stations.

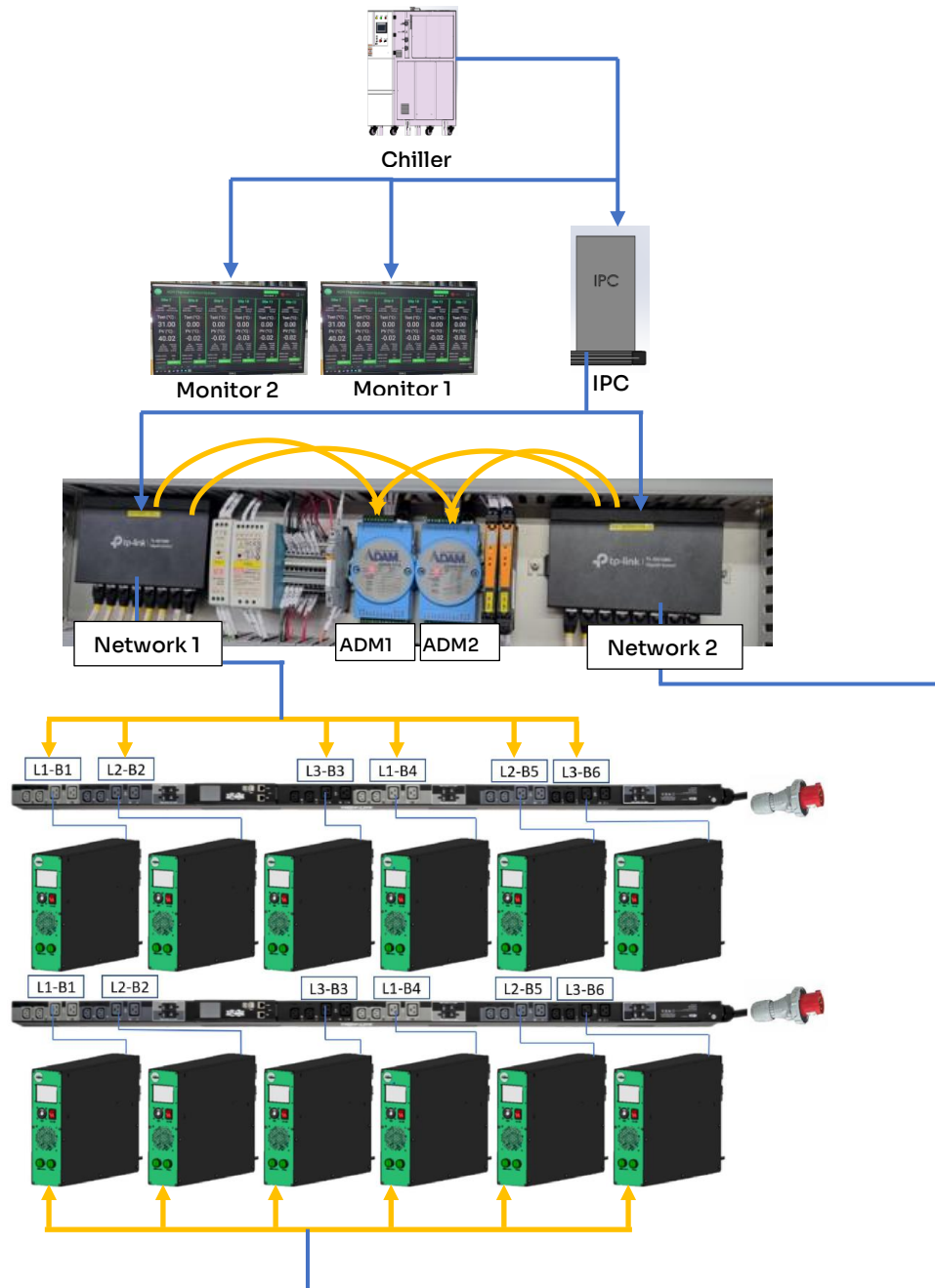


Figure 5-1 Flowchart of cable using ATC

5.1.1 ATC LCD

ATC LCD will display all the actual and current force value for each respective controllers.



Figure 5-2 ATV LCD screen



Table 5-1 ATV LCD icon function and description

Index	Function	Definitions
1	Ready	Ready state is ON when there is no Error.
2	Die Force (Actual) Die Force (Desired)	Current force on Die Actual Die force set by User
3	Cylinder Force (Actual) Cylinder Force (Desired)	Current force on Cylinder Actual Cylinder force set by User
4	Heatlet Zone Info	Set Temperature: Desire Temperature set by User (°C) Process Value: Thermal Head Temperature (°C) Mode: Following are the system mode: • Heater Following mode (HTF) • DUT Following Mode (DTF) Heater Power: Power applied on Thermal Head Liquid Valve: Value of liquid flowrate from chiller
5	Alarm Description	Display errors occur in ATC system



5.2 User Interface

Start the application by opening:

Go to Download and click on the Retrofit folder.

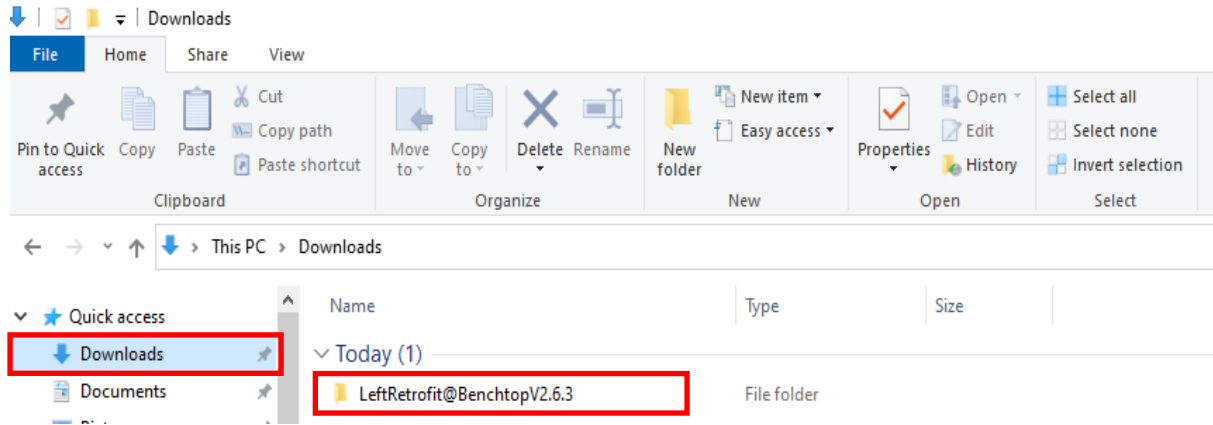


Figure 5-3 Launched Thermal Control System

Start the application by opening:

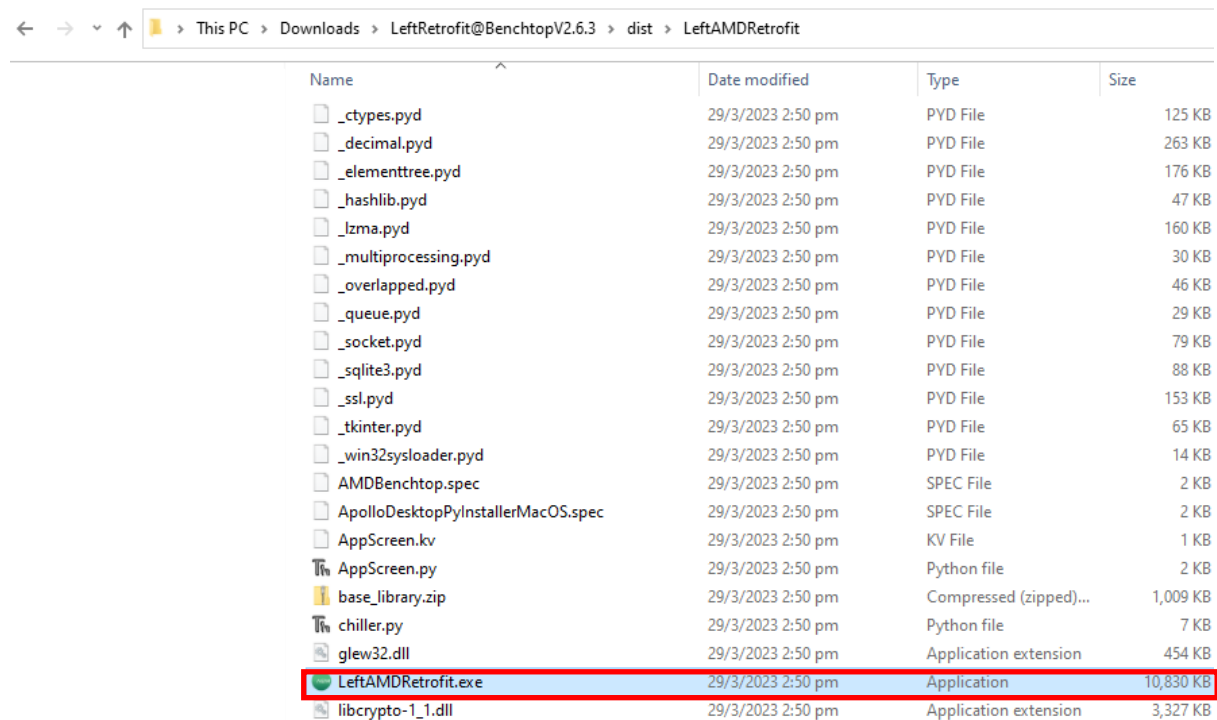


Figure 5-4 “LeftAMD Retrofit.exe” in the folder [LeftRetrofit@BenchtopV2.6.3](#)

5.3 GUI for AEM 1kW Thermal Retrofit System

5.3.1 Common Toolbars

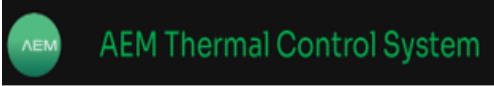
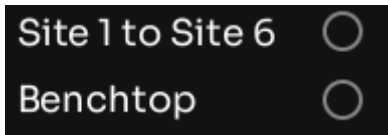
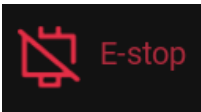
Top Toolbar

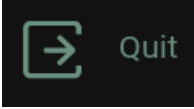
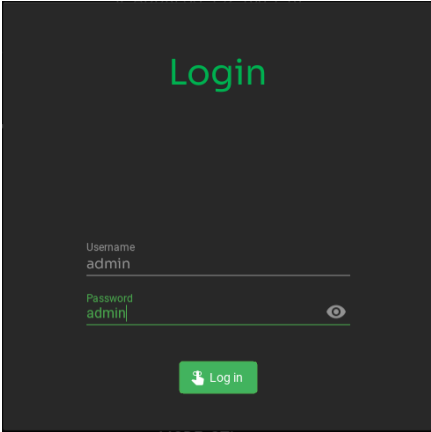


Figure 5-5

The toolbar above is the topmost label that displays peripheral connection statuses. SW automatically connects to peripherals during the launch. If a peripheral is not connected after the launch, user may retry by clicking the 'Benchtop' checkbox button.

Table 5-2

GUI Name / Illustration	Description
TITLE DISPLAY 	AEM Thermal Control System is a static title assigned in the topmost label which will be in every screen of the UI. It displays the title of the project.
MODE TAB  	Mode in here allows user to switch from Operator Mode to Engineering mode. All the controls of the parameter will be enabled only in Engineering mode. Operator mode only for view the response of the systems from the dashboard screen but not to change the variables. Default Mode – Operator Mode.
PROJECT SWITCH TAB 	Enable user to use single GUI for multiple projects. User can select the number of stations need to be connected by selecting the project name. Site 1 to Site 6 – 12 Station System Benchtop – 3 Station System
E-STOP TAB 	Emergency Stop Button tend to cut off heater powers across all zones, stop running scripts including test script and FPGA scripts. All the process will be cut-off immediately when this emergency button is clicked.

GUI Name / Illustration	Description
QUIT TAB 	Quit Button allows users to exit the GUI but all running processes will be continued. User can reactivate the GUI and continue the testing. Useful if the GUI hangs.
POP-UP WINDOW – LOGIN 	This window will only be visible when user click the Operator Mode tab change to Engineer Mode.

Bottom Toolbar

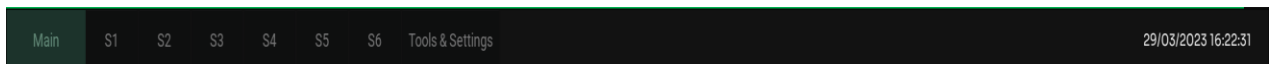
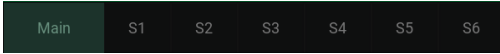
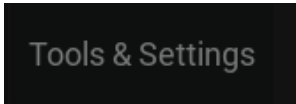
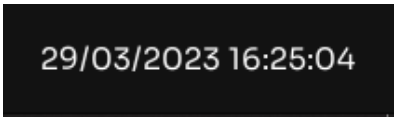


Figure 5-6

The toolbar above is a common toolbar which is located at the bottom of the GUI window. In this window, user can switch from main screen to individual station screen to control the variables. Current date and time will be always displayed at the right bottom corner of the GUI.



Table 5-3

GUI Name / Illustration	Description
SCREEN TAB 	This tab allows operators to switch screen from Main Screen to Individual Station Screen. 3 individual screens are presented in Benchtop GUI. S1 = Station 1 S2 = Station 2 S3 = Station 3 S4 = Station 4 S5 = Station 5 S6 = Station 6
TOOLS & SETTINGS TAB 	This tab allows engineers to do settings for the thermal head. Not for operator usage.
DATE & TIME DISPLAY 	Live date and time will be displayed here.



5.3.2 Main Screen

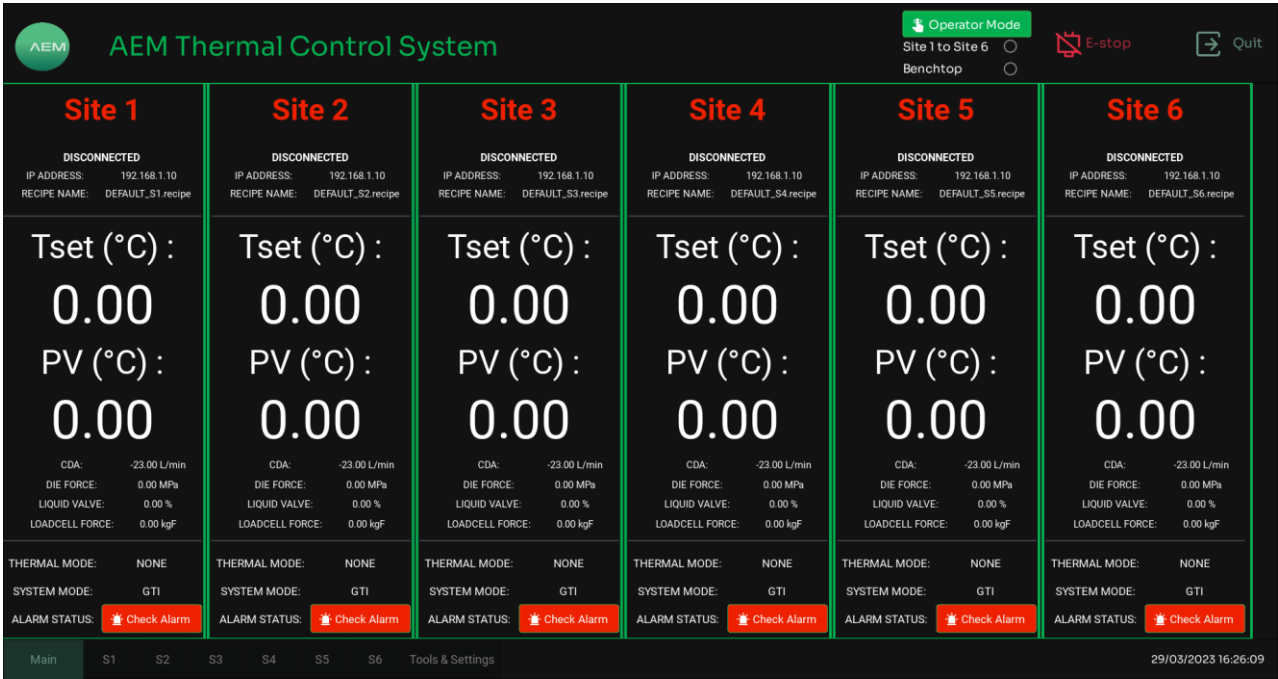


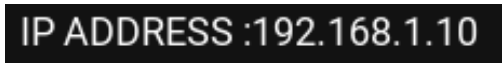
Figure 5-7

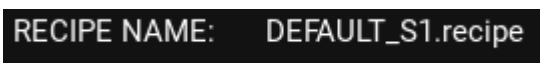
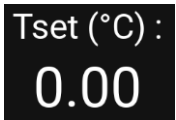
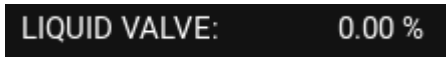
The Main Screen of Benchtop GUI displays the 3 station systems and its responses. When a user clicks the Benchtop check button at top tool bar, all 3 stations will be automatically connected and visible in connection display.

Table 5-4

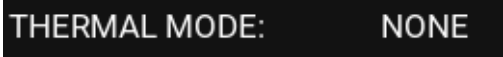
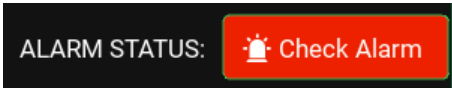
GUI Name / Illustration	Description
COLUMN	Each column consists of individual station details.



GUI Name / Illustration	Description												
SITE ID 	Topmost label in the column that indicates the Site ID. Orders of Site IDs are static. Site ID will turn into GREEN highlight when there is no alarm or GUI connects with the relevant site. RED highlight indicates presence of alarm or disconnection from the ATC system.												
CONNECTED / DISCONNECTED 	Connected / Disconnected displays the communication between SW and ATC in the system. Connection will be established automatically when physical connections are assigned. Users can't make any changes here, only for display. Users can manually change the state of this UI button in individual screen.												
IP ADDRESS 	IP address displayed here is based on user's setting in Tools & Settings (IP Config screen). Default IP addresses for Benchtop are : Site 1: 192.168.1.10 Site 2: 192.168.1.20 Site 3: 192.168.1.30 Default IP addresses for Retrofit are : <table><tr><td>Site 1: 192.168.1.21</td><td>Site 7: 192.168.1.27</td></tr><tr><td>Site 2: 192.168.1.22</td><td>Site 8: 192.168.1.28</td></tr><tr><td>Site 3: 192.168.1.23</td><td>Site 9: 192.168.1.29</td></tr><tr><td>Site 4: 192.168.1.24</td><td>Site 10: 192.168.1.30</td></tr><tr><td>Site 5: 192.168.1.25</td><td>Site 11: 192.168.1.31</td></tr><tr><td>Site 6: 192.168.1.26</td><td>Site 12: 192.168.1.32</td></tr></table>	Site 1: 192.168.1.21	Site 7: 192.168.1.27	Site 2: 192.168.1.22	Site 8: 192.168.1.28	Site 3: 192.168.1.23	Site 9: 192.168.1.29	Site 4: 192.168.1.24	Site 10: 192.168.1.30	Site 5: 192.168.1.25	Site 11: 192.168.1.31	Site 6: 192.168.1.26	Site 12: 192.168.1.32
Site 1: 192.168.1.21	Site 7: 192.168.1.27												
Site 2: 192.168.1.22	Site 8: 192.168.1.28												
Site 3: 192.168.1.23	Site 9: 192.168.1.29												
Site 4: 192.168.1.24	Site 10: 192.168.1.30												
Site 5: 192.168.1.25	Site 11: 192.168.1.31												
Site 6: 192.168.1.26	Site 12: 192.168.1.32												

GUI Name / Illustration	Description
RECIPE NAME 	Recipe set in the individual recipe screen will be displayed in this label. Users can't make any changes in this main screen but can ensure that the set recipe is the preferable one.
SET TEMPERATURE 	Tset is the temperature set by user in recipe screen or TJset value receiving in GPUTJ mode. This will be in large scale to ensure higher visibility to the users. Degree Celsius (°C) is the unit measured in for the set temperature.
PROCESSED VALUE 	PV is the process value reads from the heater which responds to the set temperature setted by user. GUI will receive this process value through ATC system which is measured in Degree Celsius (°C). PV in GUI should be correlated to the PV in ATC controller system.
COMPRESSED DRY AIR 	Indicates the actual incoming dry air for each individual station which measured in Liter per minute.
DIE FORCE 	Indicate the pressure value applied on the Thermal Head
LIQUID VALVE 	Indicate the percentage of liquid flowrate from chiller
LOADCELL FORCE 	Loadcell Force is measuring weight as a change in pressure of the internal filling fluid
SYSTEM MODE 	There are 2 system modes for the benchtop / retrofit thermal system, GTI mode and AEM mode. Mode will be displayed in this label based on the system operation. Default mode: GTI.



GUI Name / Illustration	Description
	Users can't change the mode here (dashboard screen), only changeable in individual site screen.
THERMAL MODE 	Under GTI mode, there are 3 thermal modes; Tcase, TDiode and SBTSl. Under AEM mode, there are 2 thermal modes; HTF and DTF. Primary modes will be received from the GTI mode. RTD temperature control mode (HTF and DTF) works in AEM mode only.
ALARM STATUS 	Alarm section lists out the alarm alerts user about 'Force NG' by showing RED indicator. It occurs when the set force for engage is low. Over temperature indicates the read temperature exceed the cut-off temperature. HV OFF will be RED when high voltage is inactive. GREEN and system healthy indicate there's no alarm activated, and system is ready to initiate the process.

5.3.3 Common Toolbars On Individual Station Screen

a. TOP TOOLBAR



Figure 5-8 Top label of individual station screen

The toolbar above is located below the top toolbar of the Main Screen. It is static for all the individual station screens.

Table 5-5

GUI Name / Illustration	Description
STATION ID 	First label that indicates the station ID. User can't change the station ID, only for display.
CONNECTED / DISCONNECTED CONTROL 	Connected / disconnected button here is controllable and user can manually connect to the ATC system by clicking this button. GREEN BUTTON indicates CONNECTED RED BUTTON indicates DISCONNECTED
IP ADDRESS SETUP 	User can setup their preferred IP address here and press the connection button to establish connection to the station.

b. LEFT TOOLBAR

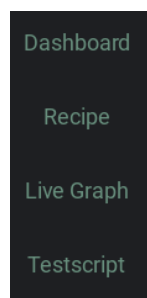


Figure 5-9 Menu of Left Toolbar

The toolbar above is located at the top left of the individual station screens. It is static for all individual station screens. This is the directory that will lead to dashboard, recipe, live graph, and test script screens.

5.3.4 Individual Station – Dashboard Screen

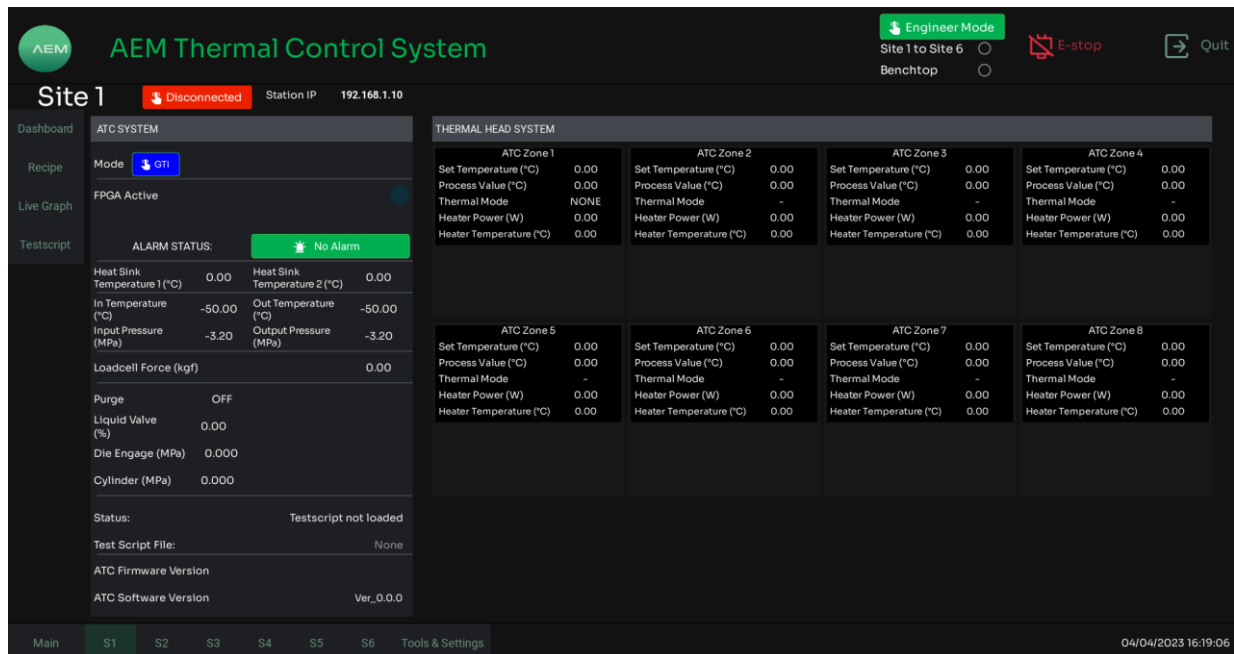
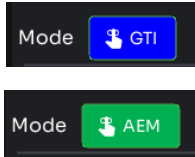
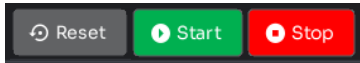


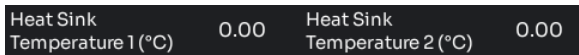
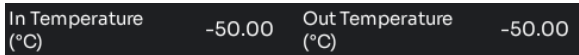
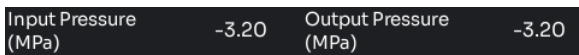
Figure 5-10

Individual Station Dashboard Screen is designed to be the main display and control screen for the users. User performs essential controls and read on this screen. It allows users to read ATC parameters and statuses.

Table 5-6

GUI Name / Illustration	Description
SYSTEM MODE CONTROL 	There are two modes which are GTI and AEM. GTI mode where command will be received from Master Controller and user is not allowed to take control of the parameters. In AEM mode, user is allowed to set the parameters and read the response by their preference. Default: GTI mode.
SYSTEM OPERATION CONTROL 	Buttons in system section allow users to reset, start and stop the running script. Green indication shows its active. Activation and deactivation of these button will be reflected in the Main Screen System Status display.

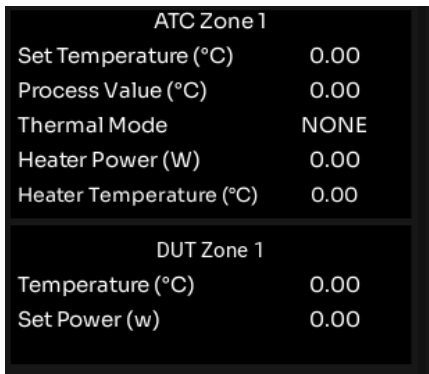


GUI Name / Illustration	Description
FPGA ACTIVE 	FPGA Active section enables users to identify FPGA status via GREEN light indication when its active.
ALARMS 	Alarms section alerts users in Red when any abnormalities occur. GREEN and alarm status in 'NO ALARM' indicates system is healthy and ready to initiate the process. 8 type of alarm code will be presented in the check alarm pop up window. There are: • FPGA High Voltage is Turn Off condition • Temperature is too high • Die Force is not engaged • Low CDA flow Rate • Ethylene Glycol Leak Detected • Timeout occur and switch to IDLE mode • Loadcell Force <30kgf and heatsink temperature <25°C • Chiller Alarm triggered.
HEATSINK TEMPERATURE 	Cold plate temperature displays the internal temperature of the heater.
TEMPERATURE 	Inlet and Outlet temperature of chiller fluid in Degree Celcius.
INPUT & OUTPUT PRESSURE 	Input and Output of chiller fluid in MPa
LOADCELL FORCE 	Actual pressure applied on DUT measured in kgf



GUI Name / Illustration	Description
PURGE 	User can enable/disable dry air purge via UI by clicking ON/OFF buttons in this section. GREEN light indicates the button is activate.
LIQUID VALVE 	User can control the liquid valve flow rate at this section. Auto mode allows the flow rate to be function in dynamic way. Manual allows users to set specific flow rate and maintain it throughout the process. Users can set the flow rate in manual mode in recipe screen.
DIE ENGAGE 	DIE can be attached and detached via UI by clicking Engage/Disengage button in this section. It also displays the force of engaging in MPa for Diaphragm Engage. GREEN light indicates the button is activated.
CYLINDER 	Display measured force of cylinder engaged controlled by handler.
TEST SCRIPT STATUS 	ATC station connection status can be seen in this section. Loaded: Test script opened & loaded in 'Test Script' screen, yet script not executed Running: Test script executed, system operating Stop: Test script execution stopped Complete: Execution complete Disconnect: Execution interrupt by a stop
TEST SCRIPT FILE 	Reflects the file name opened & loaded from the 'Test Script' screen.



GUI Name / Illustration	Description
VERSION DISPLAY 	Firmware Version: Shows Micro Controller Firmware Version of the controller box. Software Version: Micro Controller Software version. User should ensure all controllers have same firmware and software version.
THERMAL HEAD 	ATC ZONE 1-8 Displays the measurement of Measured Temperature (°C), Power (W), Mode and Set Temperature (°C) from ATC for 1 zone in Thermal Head. The preferred values can be set in 'Test script' screen. These data interpret as line graph in real time in 'Live Graph' screen. Only 1 zone is enabled in benchtop system. ATC Zone 1 = Heater Channel 1 = Enabled ATC Zone 2 = Heater Channel 2 = Disabled ATC Zone 3 = Heater Channel 3 = Disabled ATC Zone 4 = Heater Channel 4 = Disabled ATC Zone 5 = Heater Channel 5 = Disabled ATC Zone 6 = Heater Channel 6 = Disabled ATC Zone 7 = Heater Channel 7 = Disabled ATC Zone 8 = Heater Channel 8 = Disabled DUT ZONE 1-8



GUI Name / Illustration	Description
	Displays the measurement of Temperature (°C) & Set Power (W) from read from DUT for 1 zone. The preferred values set in 'TTV Recipe' screen. These data interpret as line graph in real time in 'TTV Graph' screen. DUT Zone 1 = Heater Channel 1 = Enabled DUT Zone 2 = Heater Channel 2 = Disabled DUT Zone 3 = Heater Channel 3 = Disabled DUT Zone 4 = Heater Channel 4 = Disabled DUT Zone 5 = Heater Channel 5 = Disabled DUT Zone 6 = Heater Channel 6 = Disabled DUT Zone 7 = Heater Channel 7 = Disabled DUT Zone 8 = Heater Channel 8 = Disabled

5.3.5 Individual Station – Recipe Screen

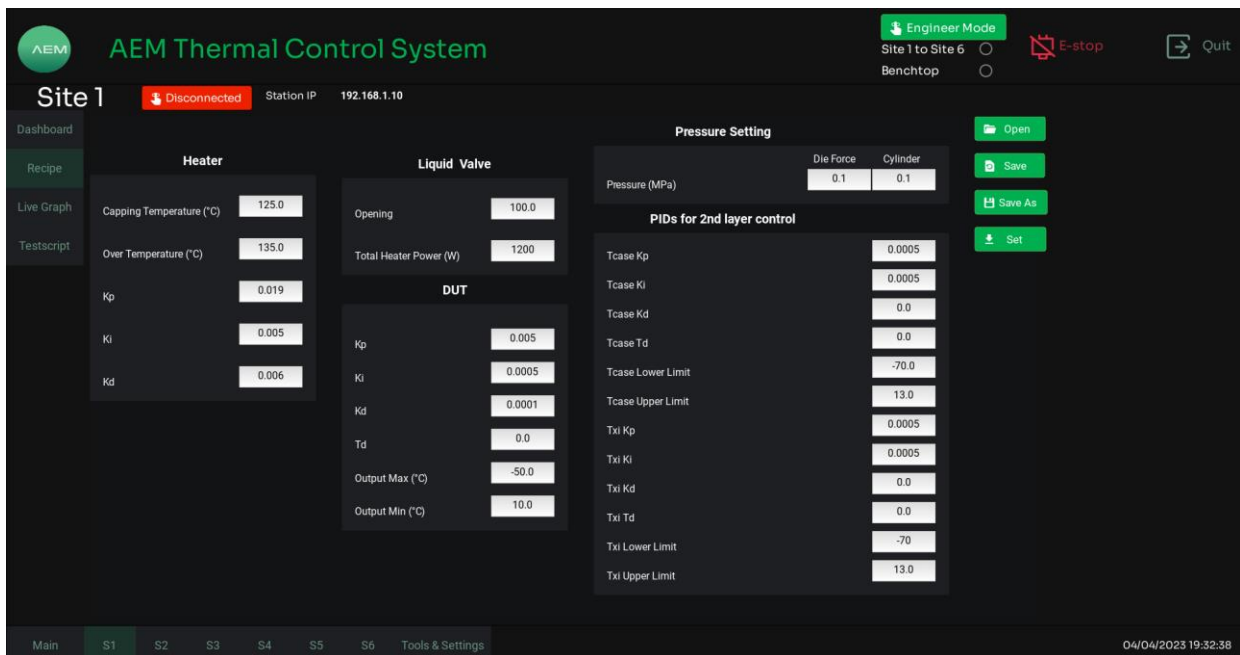
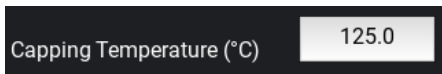
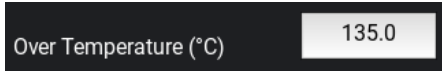


Figure 5-11

Individual Station Recipe Screen is the controller screen for the system. It contains 6 sections which in specific are: Heater, Liquid Valve, DUT, Pressure Setting, PIDs for 2nd Layer Control and Control Buttons.

Heater

Table 5-7

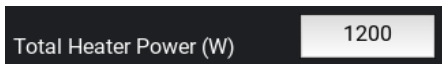
GUI Name / Illustration	Description
CAPPING TEMPERATURE 	Set the maximum limit heater temperature (°C) can reach Default value: 125.0
OVER TEMPERATURE 	Set the limit temperature (°C) to trigger thermal cut-off (TCO) Default value: 135.0
Kp 	PID Parameter (Proportional) for the Heater Heater PID value 0-1 Proportional term / Error force Default value: 0.019



GUI Name / Illustration	Description
Ki 	PID Parameter (Integral) for the Heater Sum of error over time Default value: 0.005
Kd 	PID Parameter (Derivative) for the Heater Based on rate of change of error Cause the heater to respond faster as error increases Default value: 0.006

Liquid Valve

Table 5-8

GUI Name / Illustration	Description
OPENING 	Set Range: 0.0-1.0 Adjust the liquid valve opening based on this percentage. User manually assign the opening diameter of liquid valve. Inefficient in liquid valve 'Auto' mode
TOTAL HEATER POWER 	Automatic Mode Reading from FPGA exceed the set value of Total Heater Power = Heater deactivates Reading from FPGA lower than set value in this section, Total Heater Power (W) = Heater activates back



DUT

Table 5-9

GUI Name / Illustration	Description
Kp 	PID Parameter (Proportional) Heater PID value 0-1 Proportional term / Error force
Ki 	PID Parameter (Integral) Sum of error over time Least value
Kd 	PID Parameter (Derivative) Based on rate of change of error Cause the heater to response faster as error increases
Td 	Delta Time
OUTPUT MAX 	Set maximum DUT output temperature (°C)
OUTPUT MIN 	Set minimum DUT output temperature (°C)



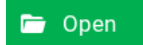
Pressure Setting

Table 5-10

GUI Name / Illustration	Description
PRESSURE 	Set the initial pressure (MPa) of diaphragm and cylinder. Die force is the centre force of diaphragm.

Control Buttons

Table 5-11

GUI Name / Illustration	Description
OPEN 	Open recipe from the folder. User can load default recipe from accessing into this button, select the preferred file and load it.
SAVE 	Save on-screen values on current file selected.
SAVE AS 	Save on-screen values to a different file as a new file.
SET 	Set the values referring to the opened file. Set the mode HTF/DTF.

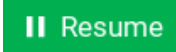
5.3.6 Individual Station – Live Graph Screen

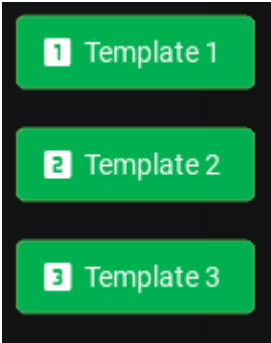
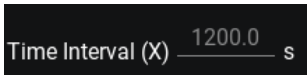
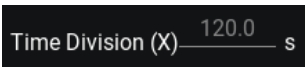
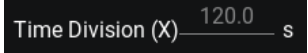
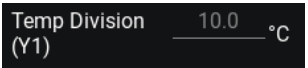
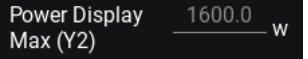
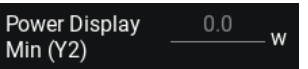
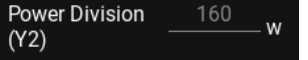


Figure 5-12

Individual Station Live Graph Screen displays real time readings in lines. Left y-axis indicates Temperature (°C) and right y-axis displays Power (W) versus x-axis that represents real-time intervals. Intervals and divisions can be modified. Legends on the graph indicates the parameters under measure with the system.

Table 5-12

GUI Name / Illustration	Description
PAUSE/RESUME  Pause  Resume	Click 'Pause' to freeze and 'Resume' to unfreeze the current view of the graph.
SAVE  Save	Click 'Save' button, goto your preferred directory, input file name & save.
CLEAR  Clear	Click 'Clear' button to remove the graph plotting on the screen.

GUI Name / Illustration	Description
TEMPLATE 	User can obtain pre set value for graph layout.
TIME INTERVAL (X) 	Users can edit & set preferred time interval at this section.
TIME DIVISION (X) 	Users can edit & set preferred time division at this section.
TEMP DISPLAY MAX (Y1) 	Users can edit & set maximum temperature to be displayed at y1-axis at this section. Y1: Left axis
TEMP DISPLAY MIN (Y1) 	Users can edit & set minimum temperature to be displayed at y1-axis at this section. Y1: Left axis
TEMP DIVISION (Y1) 	Users can edit & set temperature division at y1-axis at this section. Y1: Left axis
POWER DISPLAY MAX (Y2) 	Users can edit & set maximum power to be displayed at y2-axis at this section. Y2: Right axis
POWER DISPLAY MIN (Y2) 	Users can edit & set minimum power to be displayed at y2-axis at this section. Y2: Right axis
POWER DIVISION (Y2) 	Users can edit & set power division at y2-axis at this section. Y2: Right axis

GUI Name / Illustration	Description																																																																																																				
CHECK BUTTON <table><tr><th>ID</th><th>PV</th><th>TH</th><th>TD</th><th>TSET</th><th>PH</th><th>PD</th><th>PSET</th><th>TX</th><th>LQV</th></tr><tr><td>All</td><td>☒</td><td>☒</td><td>☐</td><td>☒</td><td>☒</td><td>☐</td><td>☐</td><td>☐</td><td>☒</td></tr><tr><td>1</td><td>☒</td><td>☒</td><td>☐</td><td>☒</td><td>☒</td><td>☐</td><td>☐</td><td>☐</td><td>☒</td></tr><tr><td>2</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☐</td><td>☒</td></tr><tr><td>3</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>4</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>5</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>6</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>7</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>8</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr></table>	ID	PV	TH	TD	TSET	PH	PD	PSET	TX	LQV	All	☒	☒	☐	☒	☒	☐	☐	☐	☒	1	☒	☒	☐	☒	☒	☐	☐	☐	☒	2	☒	☒	☒	☒	☒	☒	☒	☐	☒	3	☒	☒	☒	☒	☒	☒	☒	☒	☒	4	☒	☒	☒	☒	☒	☒	☒	☒	☒	5	☒	☒	☒	☒	☒	☒	☒	☒	☒	6	☒	☒	☒	☒	☒	☒	☒	☒	☒	7	☒	☒	☒	☒	☒	☒	☒	☒	☒	8	☒	☒	☒	☒	☒	☒	☒	☒	☒	Check and uncheck the button depending on the preference of users on graph line measurement. PV = Process Value TD = DUT Temperature TSET = Set Temperature PH = Heater Power PD = DUT Power PSET = Set Power TX = Heatsink LQV = Flow rate of liquid valve
ID	PV	TH	TD	TSET	PH	PD	PSET	TX	LQV																																																																																												
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8	☒	☒	☒	☒	☒	☒	☒	☒	☒																																																																																												

5.3.7 Individual Station – Test Script Screen

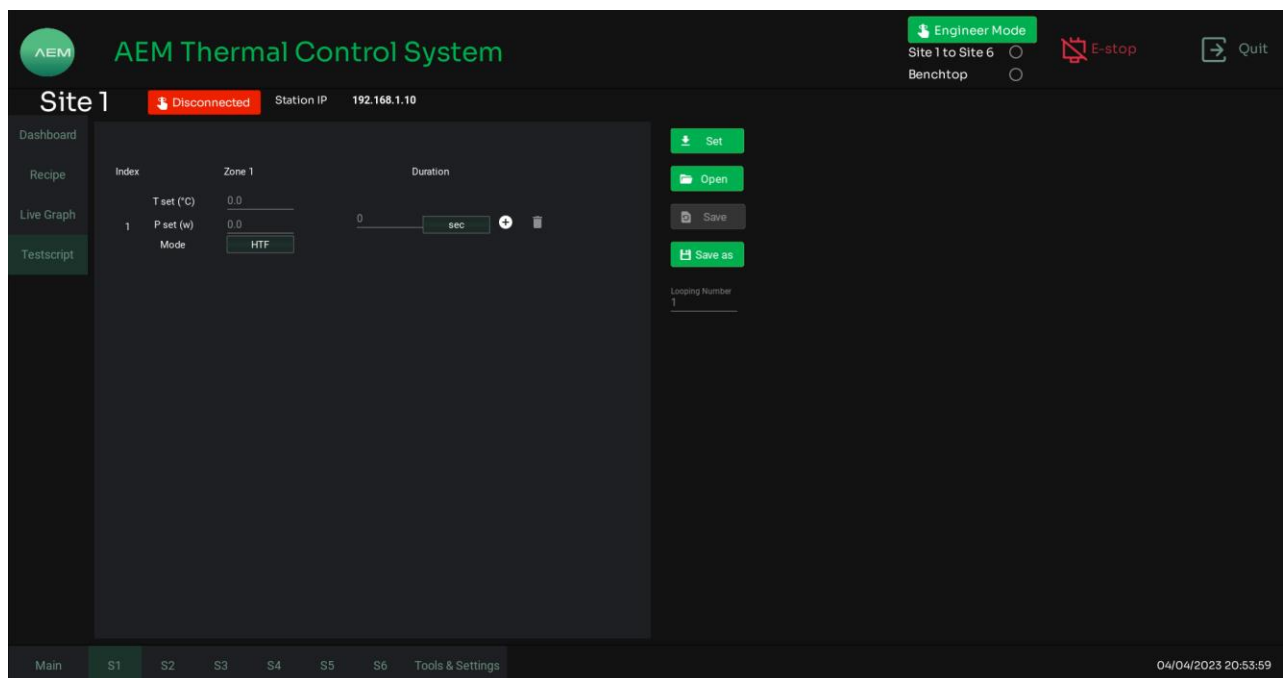
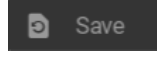
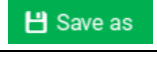
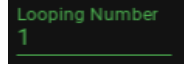


Figure 5-13

Individual Station Test Script Screen allows user to read, edit, save existing test script, or create new test script. Users can open the test script they desired and set at this screen. In 'Test Script Screen' user can set Tset and Pset values.

Table 5-13

GUI Name / Illustration	Description
TSET 	User's preferred temperature for thermal head zones can be set here. For benchtop station only 1 zone can be set the temperature and others are disabled.
PSET 	User's preferred power for the DUT zones can be set here.
DURATION 	Duration for each cycle for the set temperature and pressure.
MODE 	HTF & DTF mode can be set at here as well which aligned with the recipe screen mode.
ADD 	Add another row of data.
DELETE 	Remove entire row of data.
SET 	Load the values of Tset & Pset to the thermal head via ATC. Main screen 'Test Script Status' shows 'Loaded' if set successfully.
OPEN 	Open new test script from folder. Load back the previous / default test script.
SAVE 	Set the value to the thermal head via ATC.
SAVE AS 	Save on-screen values to a different file.
LOOPING 	Executing the same set of instructions a given number of times or until a specified result is obtained.

5.3.8 Tools & Settings - Chiller

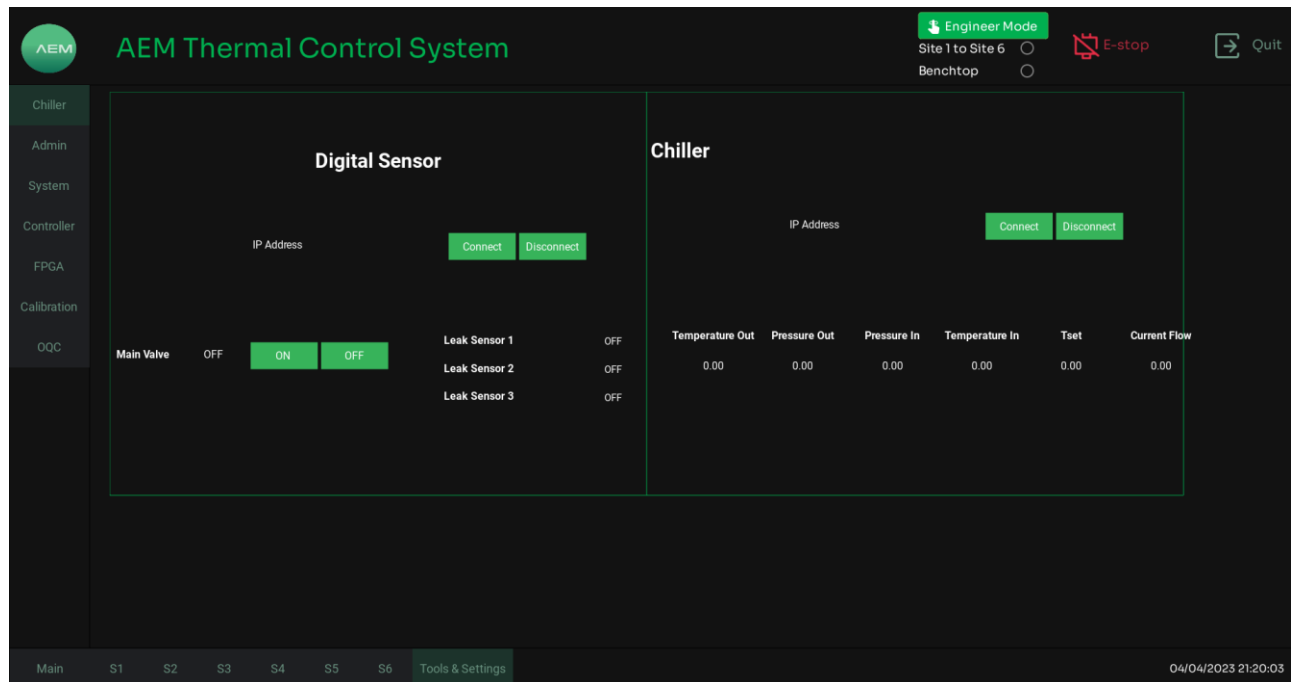


Figure 5-14

Table 5-14 DIGITAL SENSOR

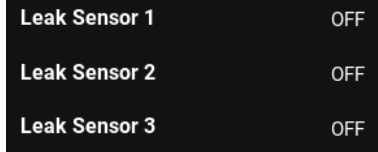
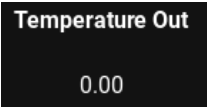
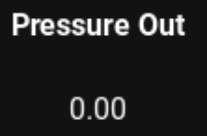
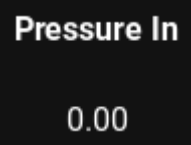
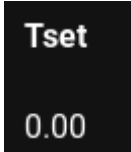
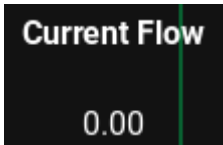
GUI Name / Illustration	Description
IP ADDRESS 	User can connect to the IP Address of leak sensor by clicking “Connect button while manually disconnect the leak sensor by clicking the button.
MAIN VALVE 	User can Turn ON or OFF chiller main valve by clicking to this button.
LEAK SENSOR 	This bar display all the sensor status of leak sensor. If the status shows ON ,leak is detected while Off means no leak is detected.



Table 5-15 CHILLER

GUI Name / Illustration	Description
IP ADDRESS 	User can connect to the IP Address of chiller by clicking “Connect button while manually disconnect the chiller by clicking the button.
TEMPERATURE OUT 	User can monitor actual temperature out from chiller.
PRESSURE OUT 	User can monitor actual pressure out from chiller.
PRESSURE IN 	User can monitor actual pressure in from chiller.
TEMPERATURE IN 	User can monitor actual temperature in from chiller.
TEMPERATURE SET 	User can monitor setting temperature of chiller.
CURRENT FLOW 	User can monitor current flowing from chiller

5.3.9 Tools & Settings – Admin Screen

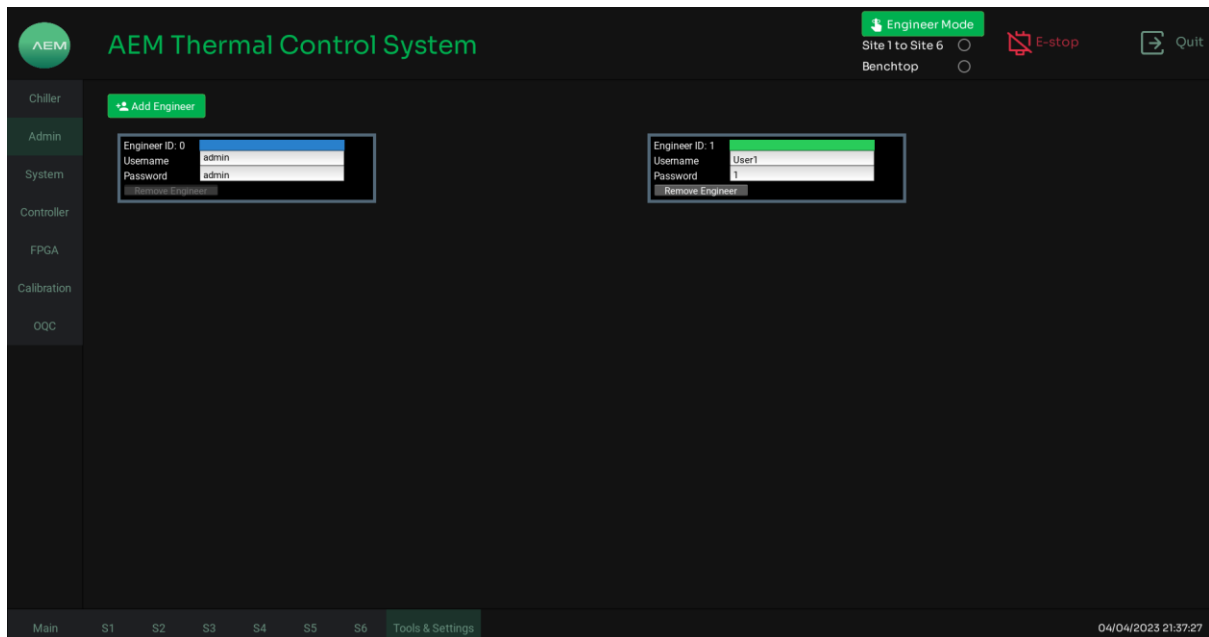
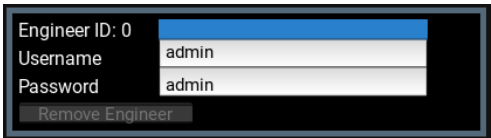
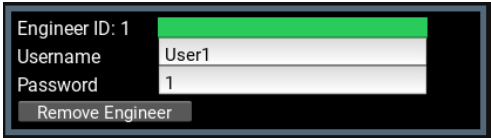


Figure 5-15

Table 5-16 Description of Admin Screen

GUI Name / Illustration	Description
ADD ENGINEER 	Adding authorised engineer who working on the UI system.
ENGINEER DETAILS ID 	Displays details of engineers added to the UI such as ID, username and password.
ACCESS ENGINEER DETAILS 	Display details of engineers who is using the UI system.
REMOVE ENGINEER 	Authorised engineer can delete out engineer not working on the system.



5.3.10 Tools & Settings – System Screen

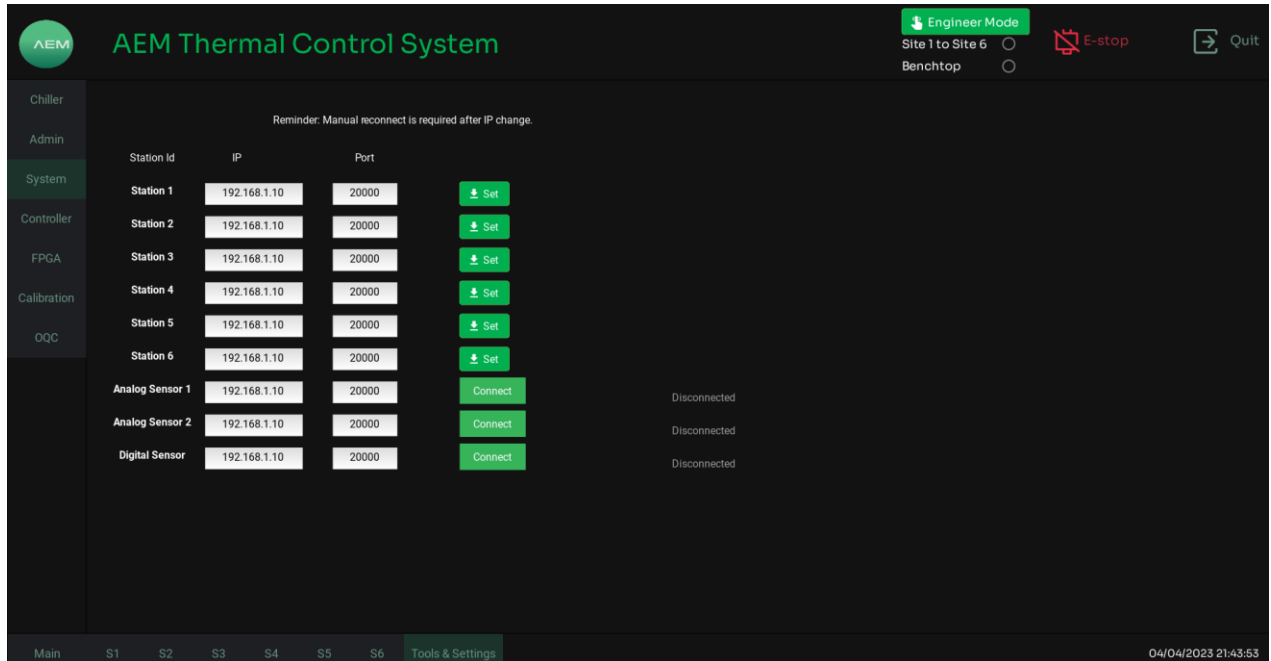


Figure 5-16

5.3.11 Tools & Settings – Controller

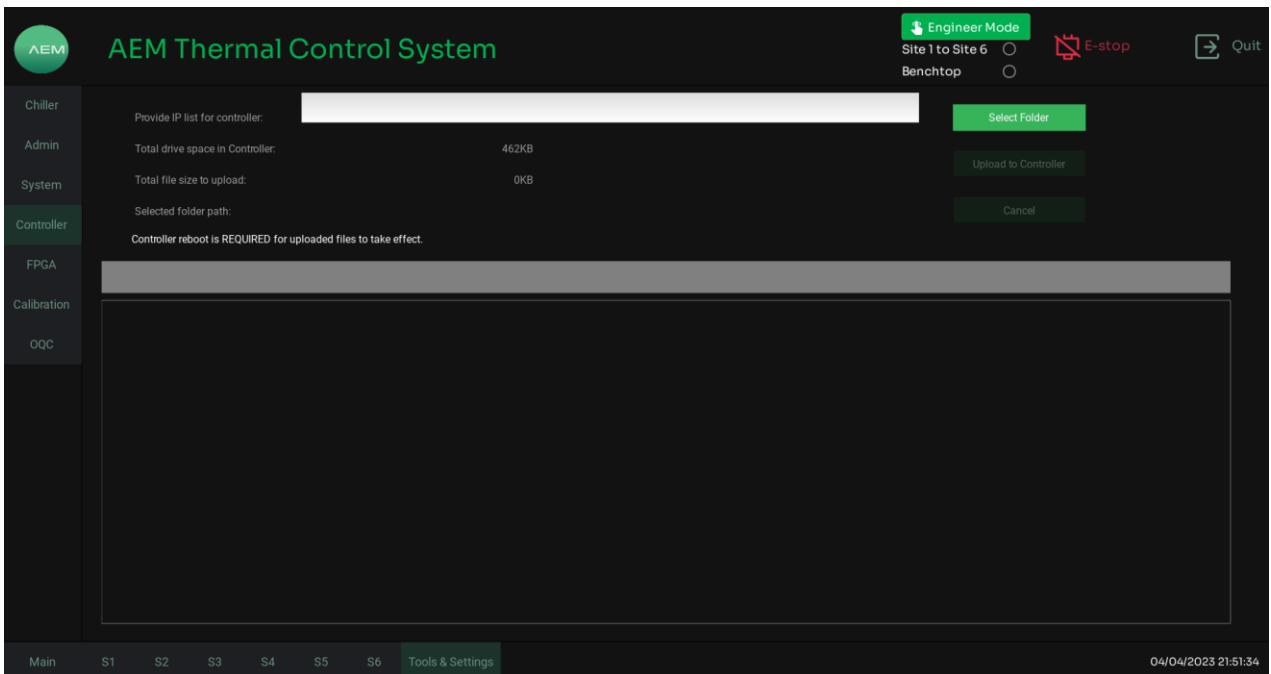


Figure 5-17



5.3.12 Tools & Settings – FPGA

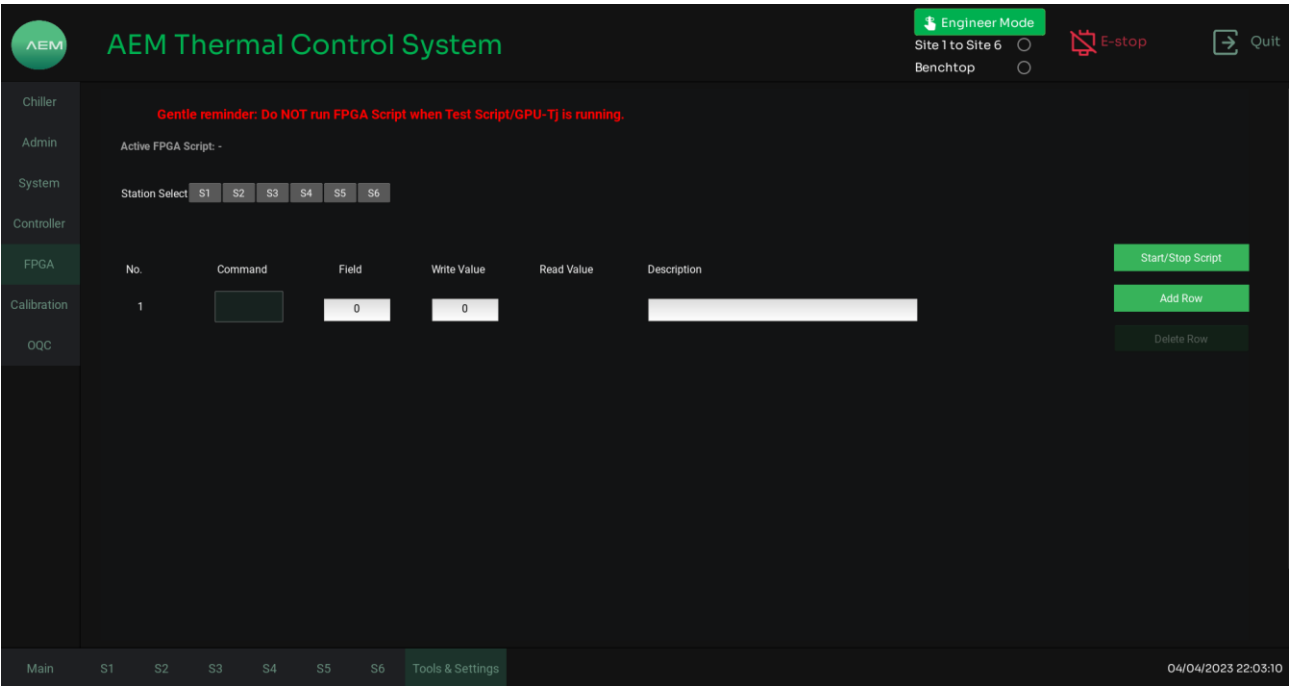


Figure 5-18 FPGA Screen

5.3.13 Tools & Settings – Calibration Screen

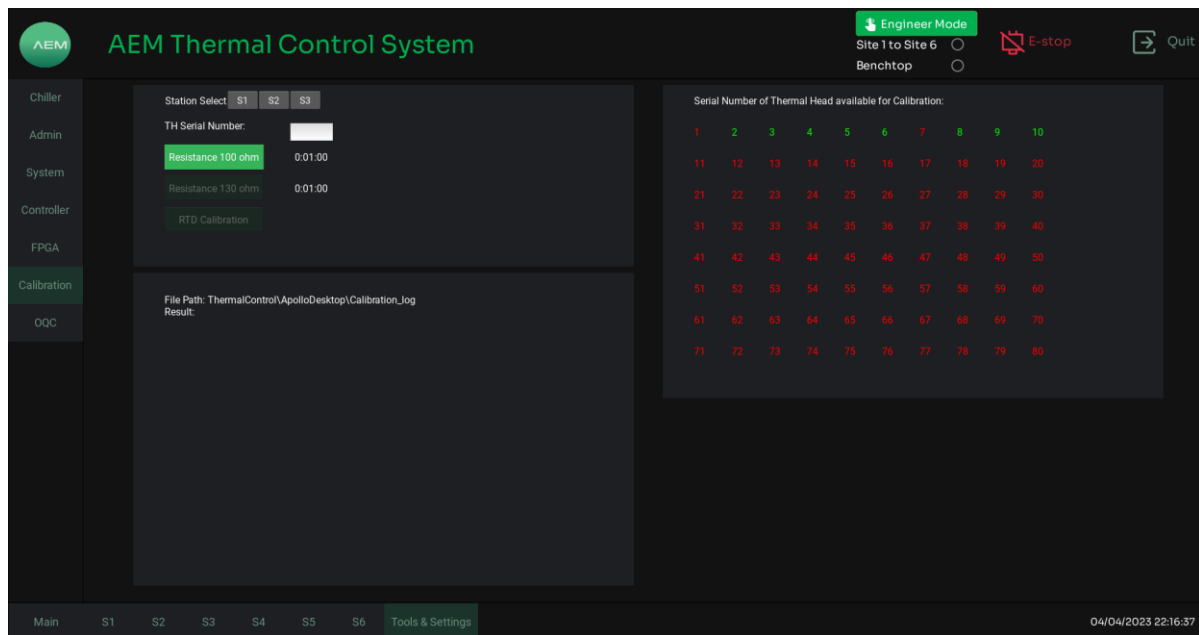
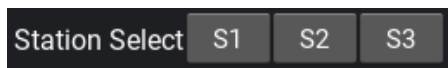
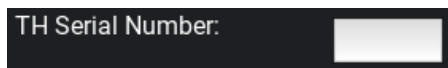
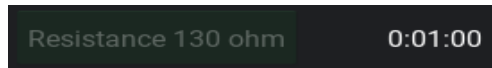


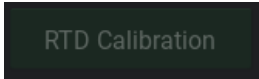
Figure 5-19

Calibration screens allow engineers to perform calibration for the thermal head. Not advisable for the users to use this screen without supervision.

Table 5-17

GUI Name / Illustration	Description
STATION SELECTION 	User able to select the station calibrating on.
SERIAL NUMBER INPUT 	Each thermal head defines for specific serial number. Engineers should enter serial number which is attached to the station at the given input section.
BUTTON TO CALCULATE RESISTIVE ERROR OF 100 OHM 	Engineers should attach the CALSET of 100 Ohm and click this button to start the calibration process. Offset value will be measured during this time. Timer will indicate the 1 minute of this process completion.
BUTTON TO CALCULATE RESISTIVE ERROR OF 130 OHM 	Once the first step and second step completed, this third button will be enabled, and engineers should attach 130 Ohm of CALSET to perform the second



GUI Name / Illustration	Description
	cycle of offset value calculation. Timer will indicate the 1 minute of this process completion.
RTD CALIBRATION BUTTON 	Last button will be enabled when the 3 steps are completed. Clicking this button will generate hex codes which will be load into the ATC system to achieve stable and reliable thermal head temperature values.

5.3.14 Tools & Setting - OQC Screen

5.3.14.1 Single Station OQC Test

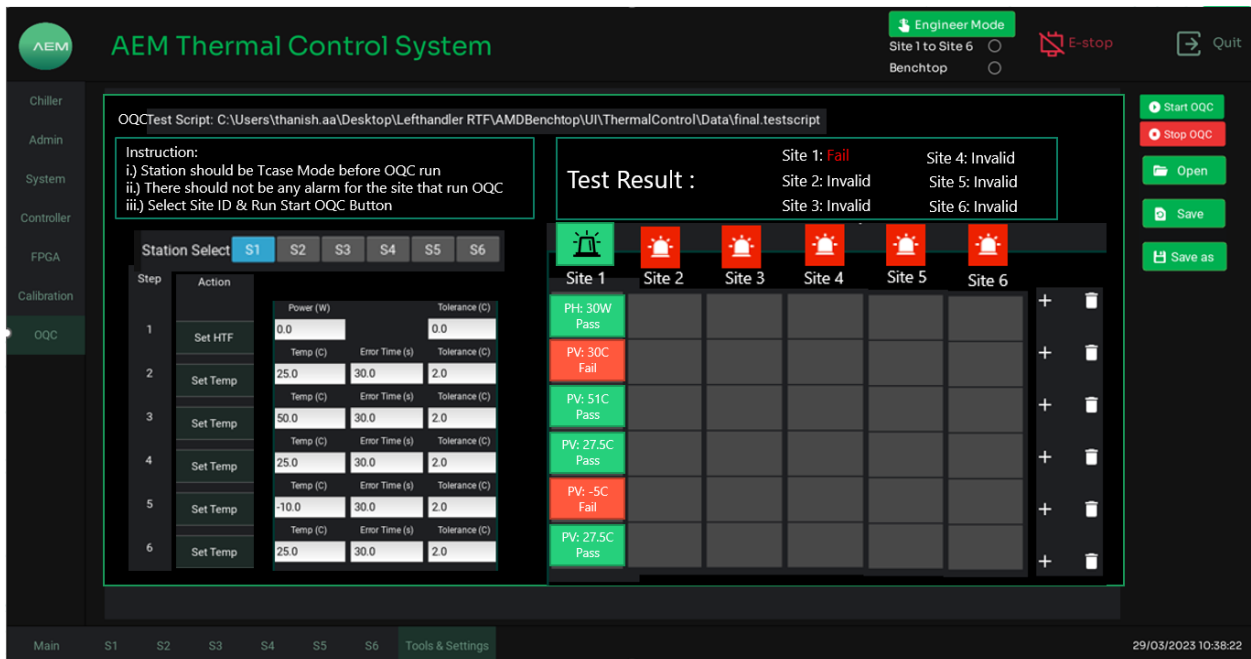


Figure 5-20

5.3.14.2 Multiple Station OQC Test

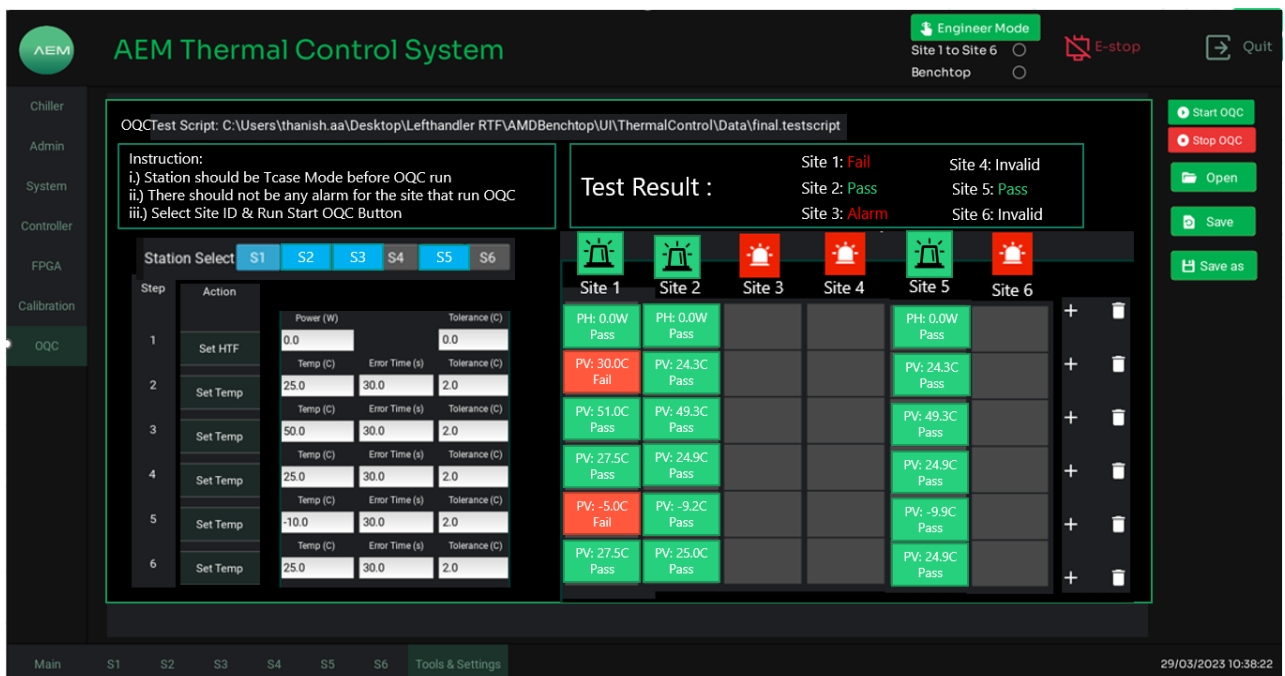
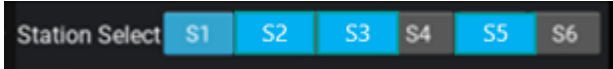
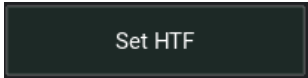
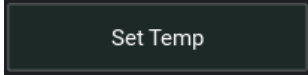
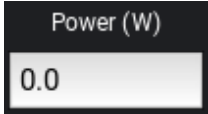
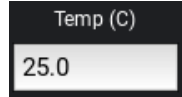
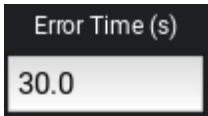
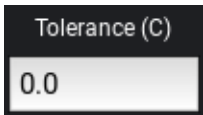
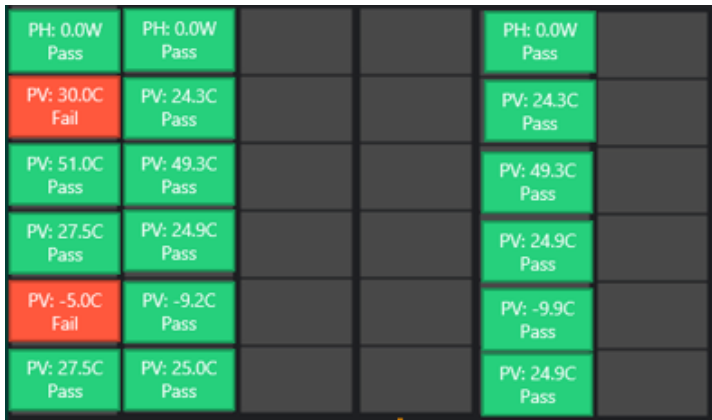
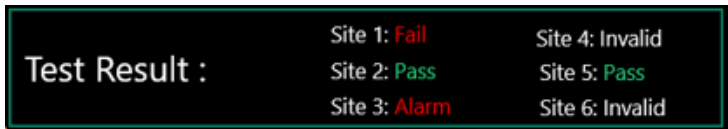
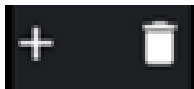


Figure 5-21

Description

Table 5-18

GUI Name / Illustration	Description
STATION SELECTION 	User can select individual station or multiple station at once to run the OQC test simultaneously
SET HTF 	User can select in the drop down box. Set HTF will allowed to set the Heater Power.
SET TEMP 	User can select in the drop down box. Set Temp to enter desired set temperature.
POWER 	User can set the power required at the box given.
TEMPERATURE 	User can set the desire temperature manually.
ERROR TIME 	User can set the parameters at the input box given here. Error time is the grace time for the actual reading match to the set temperature.
TOLERANCE 	User can set the parameters at the input box given here. Tolerance can be adjusted based on user preference.
ALARM INDICATOR 	Alarm Indicator which turn red when there's any alarm generated in the relevant site. It won't be displaying Alarm Code, but just indication whether

GUI Name / Illustration	Description
	there's alarm or not. User have to enter individual/mainscreen (Check Alarm) feature to acknowledge the alarm description. Green alarm indicates good to run OQC test.
STATUS 	Result of each set parameters will be displayed in box-tabular form. In each box, actual reading will be displayed and set the background colour green if the result passed and red for fail.
SUMMARY OF TEST RESULT 	Test Result section will display the overall result according to the sites that has been run. Even if one step fails, the overall result will be shown as fail. If there's an alarm for the site selected by the user, it will display 'Alarm'. 'Invalid' for the sites which has not been selected for the OQC test.
ADD & DELETE 	Add and Delete options are enabled for user to modify the OQC script. Scroll view will be applied if the OQC script is longer the screen size.
START OQC 	User shall click the Start OQC Button, to run the test. Liquid valve will be always in Auto Mode. It runs each steps for every station selected

GUI Name / Illustration	Description
	simultaneously. User can't start the run with any alarm triggered in the station.
STOP OQC 	Stop OQC will allow user to stop the test at any stage of the run. User have to click Start OQC to restart the test after their preferable changes.
OPEN 	Open button enable user to choose the OQC script they desire.
SAVE 	Save button enable user to save the changes which has been modified in the pre-saved script.
SAVE AS 	Save As button enable user to create a new script and save in the folder.



NOTES

[illegible]



6. Operation

6.1 Operation Overview

This section provides information on the layout of the operation panel as well as the function and location.

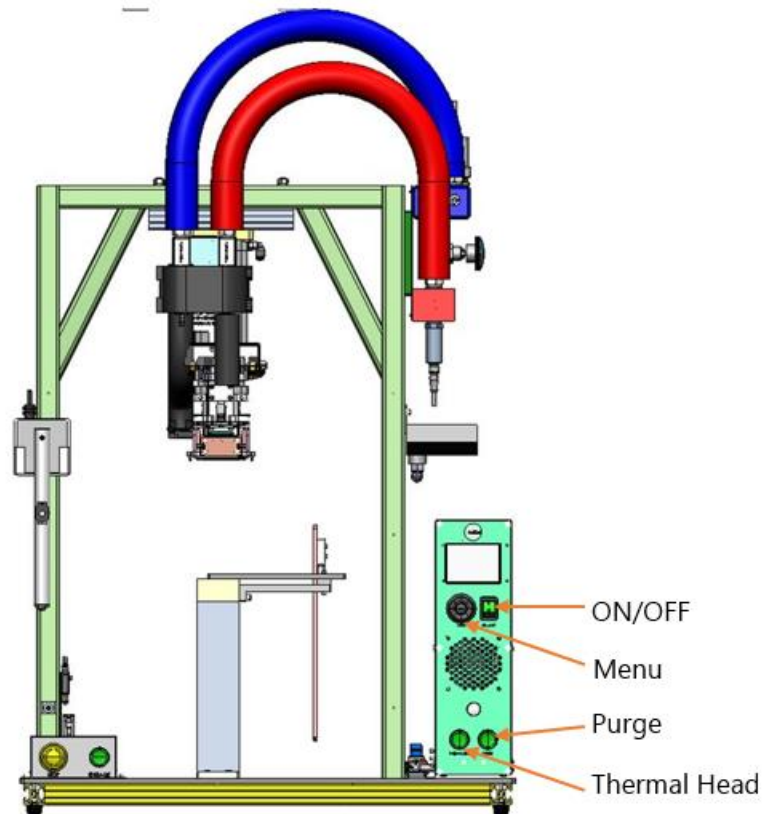


Figure 6-1

6.1.1 Push Button

Table 6-1

Push Button	Description
ON / OFF	Energizes / Deenergizes the tool
Thermal Head	Engage / Disengage the TH Total / Die force
Purge	Turns on compressed air for EGW Purging
Menu	Navigates the GUI

6.2 Machine Operating Procedure

The following descriptions state the list of hazardous energies and guide the user in step-by-step procedure in operating the machine.

6.2.1 List of Potentially Hazardous Energy Sources

Refer to [Section 3.1](#) for facility requirements of the AEM Thermal Station.

6.2.2 Power ON Procedure



ENSURE THAT ALL PERSONNEL ARE KEPT AWAY FROM MACHINE AS ANY MOVING PARTS CAN CAUSE SERIOUS INJURY

Check and ensure that the incoming power supply is according to the specifications provided before switching on the power. Refer to [Table 3-1](#) for correct specifications.

Switch ON the AEM Thermal Station by pressing the ON switch.

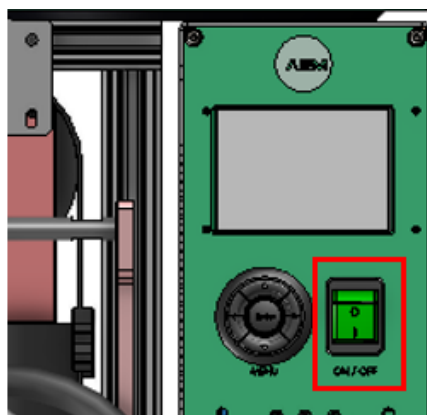


Figure 6-2 Switch ON the AEM Thermal Station

6.2.3 Power OFF Procedure

Switch OFF the AEM Thermal Station by pressing the OFF switch.

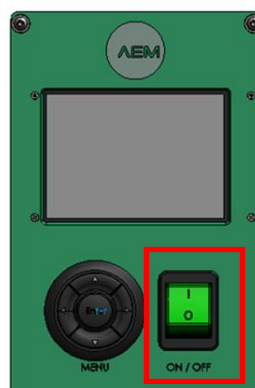


Figure 6-3 Switch OFF the AEM Thermal Station

6.2.4 Verify EGW & OFA are Connected

1. Verify that the following lines are connected to the tool:
 - EGW Supply
 - EGW Return
 - Air Supply

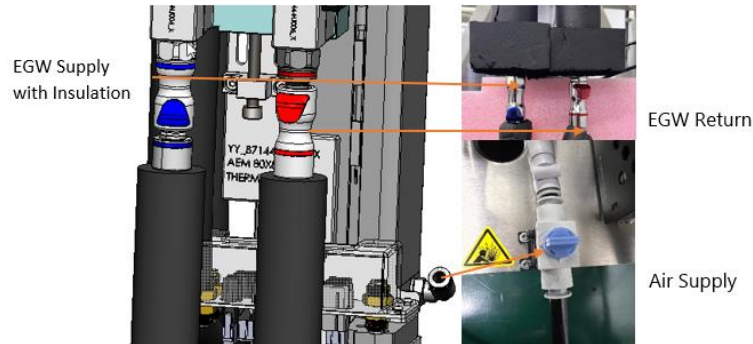


Figure 6-4 AEM Thermal Station lines

2. Make sure the EGW is circulating before conducting any testing. (**Note:** PV 1, TX1 and TX2 should be close to 10°C if the incoming EGW temperature is 10°C).
3. Check the incoming air supply by toggling the purge mode into manual. Use a hand to feel the purge air coming from the purge point at thermal head.

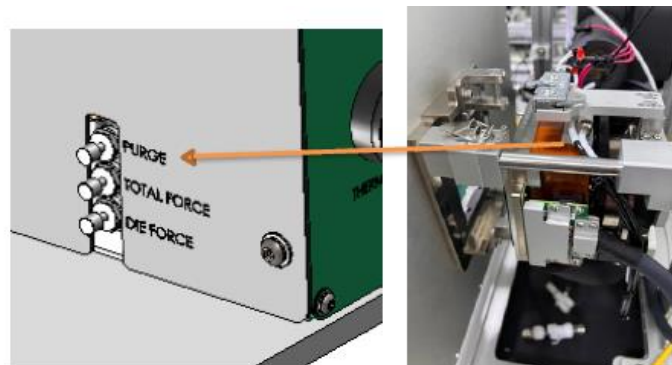


Figure 6-5

4. Check diaphragm piston functionality by pushing the Thermal Head button.



Figure 6-7 Set the target force

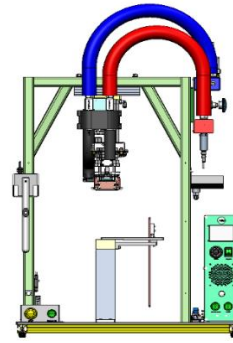


Figure 6-6 Extend and retract Thermal Head by push button



NOTES

[illegible]



7. Fault Description

7.1 Error and Handling

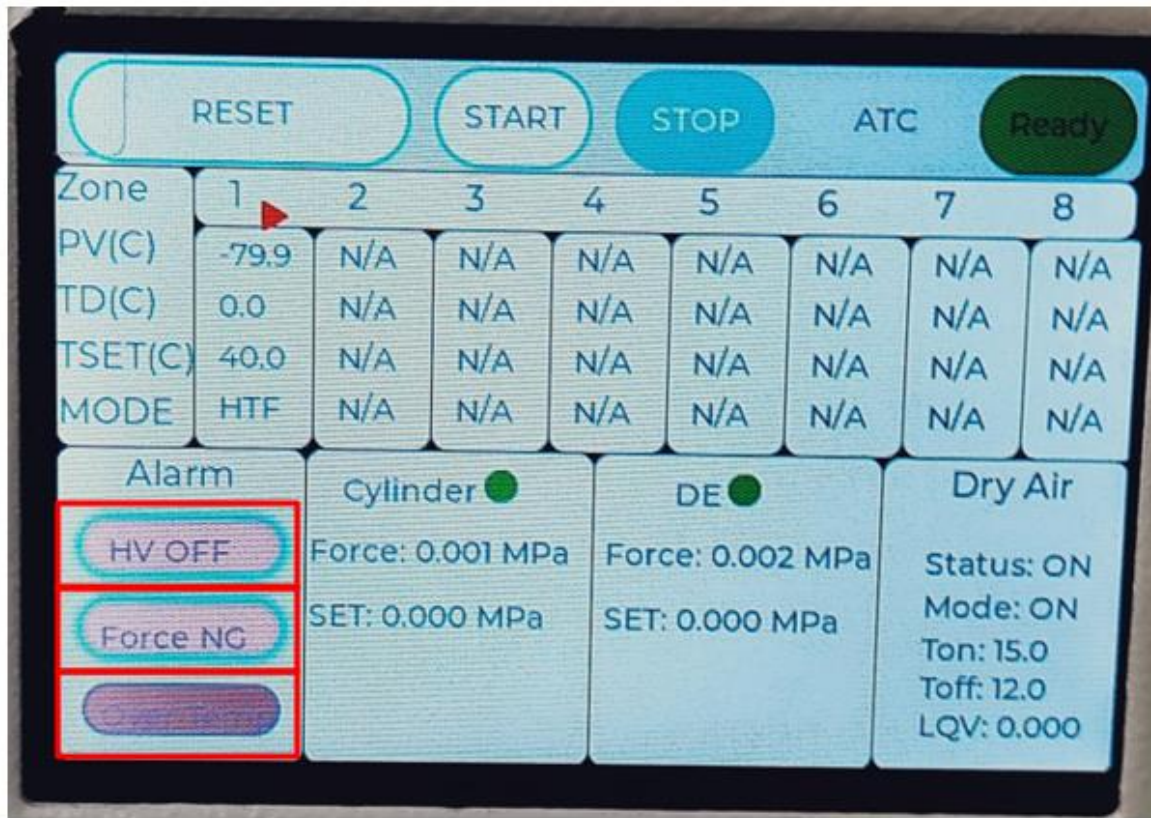
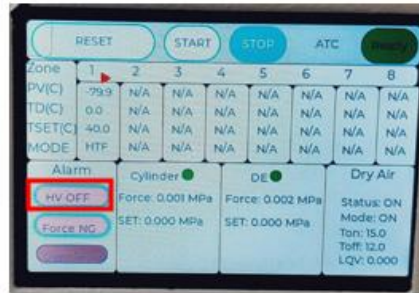


Figure 7-1

There are 3 types of error that may cause thermal control issues. The reason, effect and solution of these errors are stated in the following pages.

1. HV OFF
2. FORCE NG
3. OVERTEMP

1. HV OFF

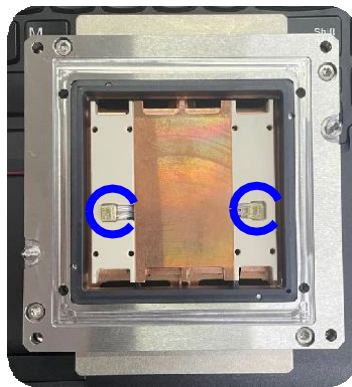


Reason:

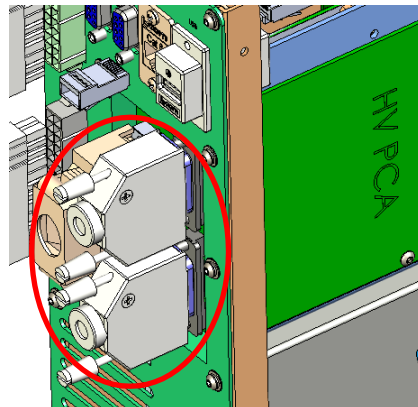
1. TCO is broken or HV board is malfunction.
2. Connector is not plugged in.
3. Interface board faulty.

Effect:

1. Zone heaters do not have power output to heat up the Thermal Heat.

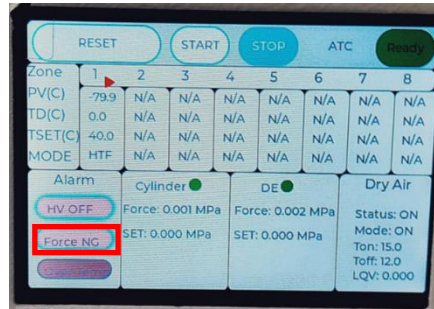


Solution:



1. Ensure TCO is not broken.
2. Connectors are plugged in.

2. Force NG



Reason

1. When Force is less than 30.0 lbf this alarm

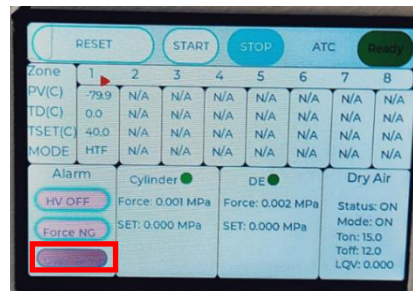
Effect:

1. Does not stop any thermal control
2. Indirect affecting thermal performance

Solution:

1. Encouraged to set Force above 30.0 lbf (Not a must for Force Setting)

3. Over Temp



Reason:

1. When temperature is above set Over Temperature Value on the recipe screen, this alarm lights up.

Effect:

1. Stop control.

Solution:

1. Ensure fluid flow is good
2. Press Reset to enable back control.



NOTES

[illegible]



8. Conversion



Mechanical Hazards: These hazards exist in any case where a component can move in such a way as to impact or otherwise injure personnel performing maintenance functions on the equipment. To avoid these hazards always operate the equipment with the safety covers in place and door interlocks enable. When performing maintenance functions with the covers open, be sure to avoid those areas where impact with the moving parts of the equipment is possible. When removing components, it may be necessary to isolate the potential mechanical energy by restraining the mechanism using cable ties or blocking.

This chapter covers all the procedures to change over the tooling to support a different test unit type. When requirements call for a change to a new package with different sizes of units, trays and/or chassis, a conversion kit must be installed. The conversion kit does not include the common part; the kit only includes the specific design parts.



ENSURE THAT A SIGN IS PLACED INDICATING THAT THE EQUIPMENT ARE BEING SERVICED OR ADJUSTED, IN ORDER TO KEEP ANY ONE FROM TOUCHING THE OPERATION PANEL.

ONLY QUALIFIED PERSONNEL SHOULD CONDUCT CONVERSION OPERATIONS.

ALWAYS SWITCH OFF MAIN POWER SUPPLY AND COMPRESSED CDA TO THE MACHINE BEFORE DOING CONVERSION OPERATION.

8.1 Thermal Head Kit Removal Procedure

1. Turn off the power supply.

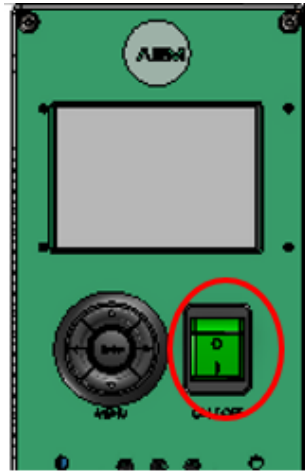


Figure 8-1

2. Disconnect the Inlet & Outlet Quick Disconnects (QDs) at the end of the thermal head hose.

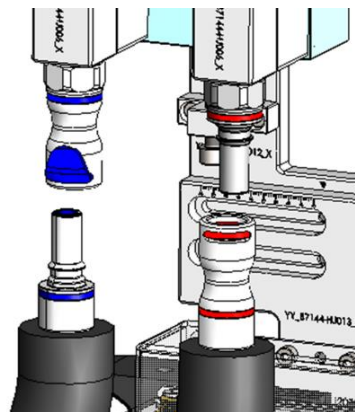


Figure 8-2

3. Disconnect the air hose and disconnect the electrical connectors.

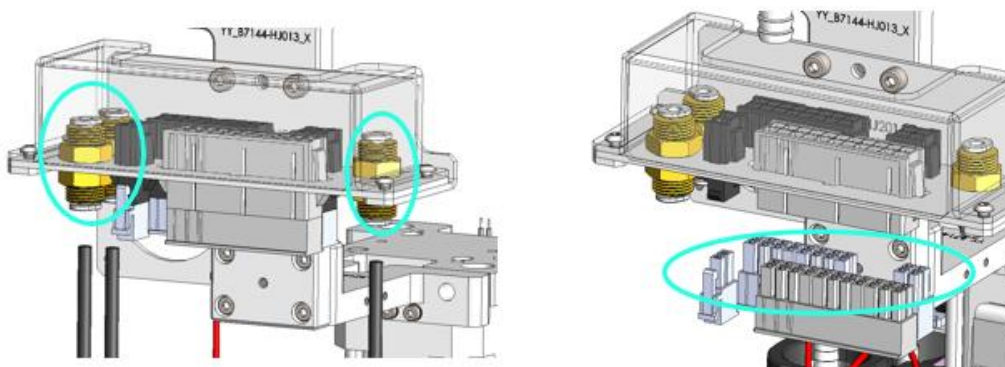


Figure 8-3

4. Catch clips on Top & Bottom of the TH are released.

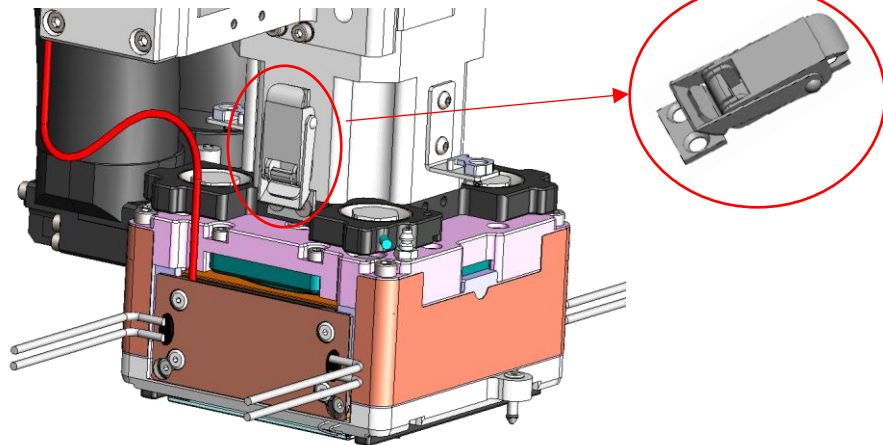


Figure 8-4

5. TH now can be removed from the bottom.

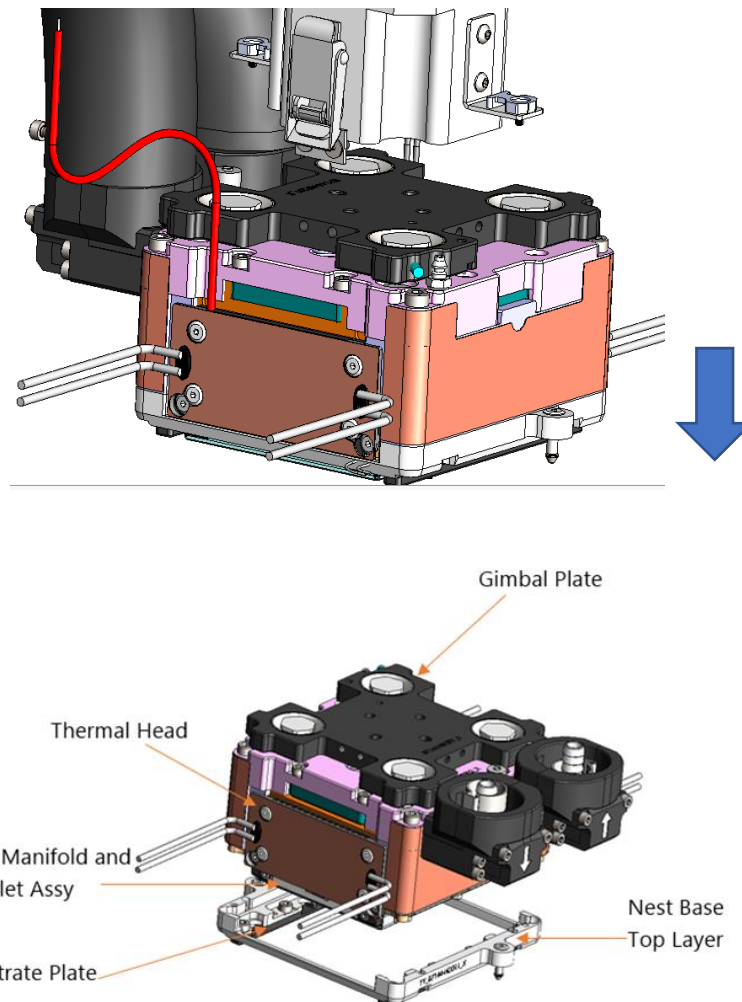


Figure 8-5

6. Remove 4x screws on kit and remove the substrate pusher plate.

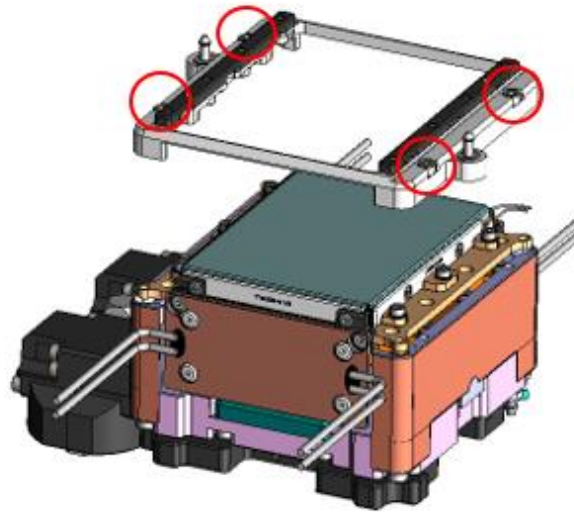


Figure 8-6

7. Kit removed from TH.

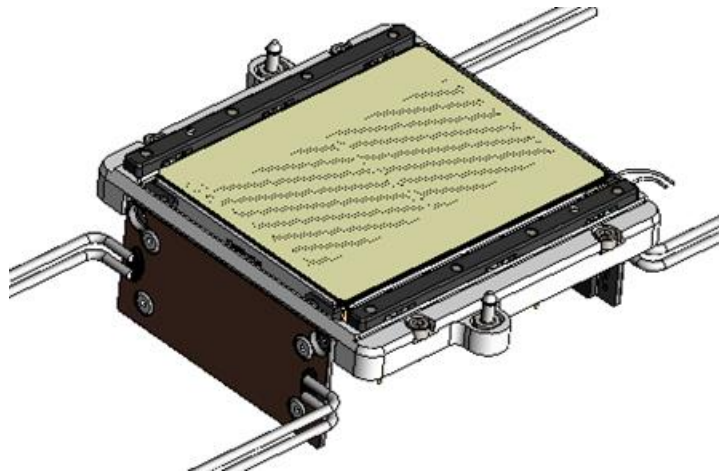


Figure 8-7

8. Change to another TH kit or TIM and reverse the step from 6 to 1.

8.2 Changing TIM

1. Remove the nest base bottom layer.

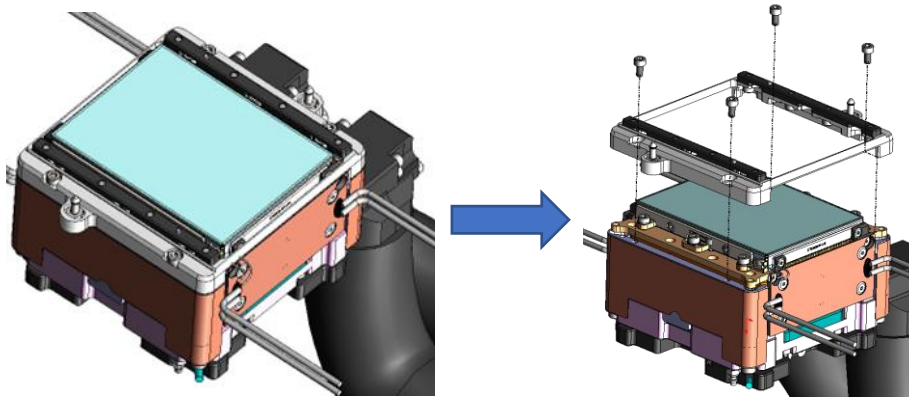


Figure 8-8

2. Remove x8 screws fastened to the TIM retainer.

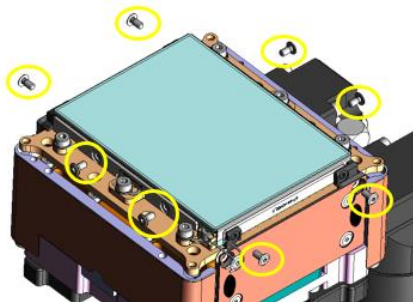


Figure 8-9

Note: Fix from the Nest Base (bottom layer) to the Substrate pusher.

3. Change TIM from the heatlet.

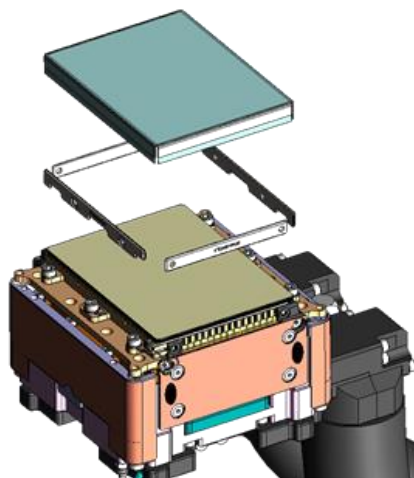


Figure 8-10



4. Reinstall the kit following the above steps in reverse.

Note: A stress test need to be conducted if new TIM is implemented.

- Make sure the station is connected to the facility (EGW and air pressure).
- Set the socket force into 140 lbf.
- Engage the socketing force to the DUT.
- Run the stress test script for 5 mins continuously (from -30°C then stay 30 sec to 110°C).
- Observe the TH and TD should be able to reach -30°C.



NOTES

[illegible]



9. Troubleshooting



9.1 Equipment Malfunction

This chapter consists of a general description of possible equipment malfunctions or failures. Each description includes the remedies or possible solution of an issue that may arise. It is set with a quick guide to assist with general ideas to overcome unwanted miss failures. We strongly recommend you to first contact the supplier for further assistance if there is any doubt on how to perform activities of troubleshooting.



ONLY QUALIFIED PERSONNEL SHOULD ATTEMPT TO TROUBLESHOOT THE DEBUG CART MACHINE.

TROUBLESHOOTING GUIDE TABLES Index of Contents:

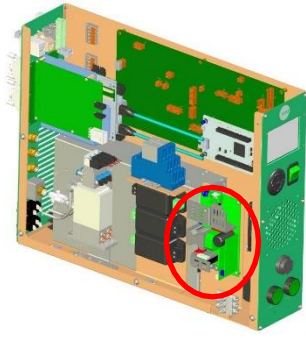
9.1 General Possible Failures: a) General machine power failures b) Over temp c) Unable to extend / retract piston diaphragm	9.2 Thermal Activation Failures: a) Temperature ramp-up cannot reach set point b) Temperature ramp-up overshoots set point
---	---



Please always follow safety guidelines, SATEFY FIRST!

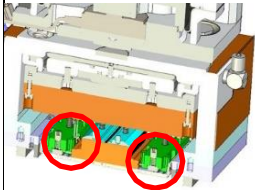
9.2 General Possible Failures

Table 9-1

Problem	Symptom(s)	Remedy / Possible Solution	Illustration / reference
a) General Machine Power Failures	Machine is SWITCHED OFF No screen display Fans are not SWITCHED ON	Check the inductor relay condition related to Main Power ON Check Incoming Power Supply	
b) Over temp	Over temp alarm Unable to start any testing Failed to maintain the temperature	Check Main Facility of EGW and ensure flow is good Check the assembly orientation. It must face up or towards pedestal	When temperature is above user set temp, this alarm lights up
c) Unable to extend/retract piston diaphragm	Cylinder not moving after toggle to extend or retract	Check Main Facility of air and ensure flow is good Check set force minimum must be above 20 lbs	

9.3 Thermal Activation Failures

Table 9-2

Problem Description	Possibility Rootcauses	Actions	Illustration / reference
a) Temperature ramp-up cannot reach set point	TIM pre-conditioning not performed	Perform stress test stated at Chapter 8	
	TIM damaged	Replace VCTIM	
		Perform stress test stated at Chapter 8	
	Thermal fuse blown	Replace thermal fuse	
	Heater damaged	Replace the thermal head kit	
b) Temperature ramp-up overshoots set point	EGW didn't turn on	Check Main Facility of EGW and ensure flow is good	
	Incorrect heater-RTD parameter	Retrieve heater-RTD parameter from record Confirm that correct parameters are displaying after programming	



NOTES

[illegible]

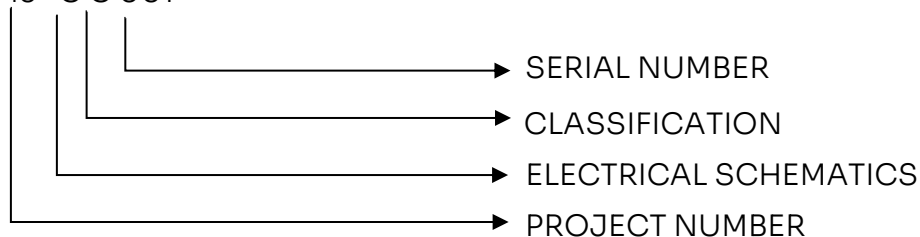


10. Electrical

10.1 Electrical Schematics Nomenclature

10.1.1 Schematics-Numbering System

B2743- C C 001



REVISION
& DATE

TITLE : TAPC 1&2 CONNECTION				 THIS IS A PROPERTY OF AEM SINGAPORE PTE LTD IT SHOULD NOT BE COPIED OR REPRODUCED WITHOUT THE WRITTEN APPROVAL OF AEM SINGAPORE PTE LTD.
DRAWING NO : B3065-CC012		REV: 0	DATE: 15/12/2019	
	INITIAL	SIGN	DATE	
DRAWN	PETHUEL ENOT		09/07/2019	
CHECKED	PETHUEL ENOT		26/10/2019	
APPROVED	ARUN KUMAR		26/10/2019	
SIZE : A3	DO NOT SCALE DRAWING	PAGE NO : 44 - 50		

10.1.2 Schematics Classification

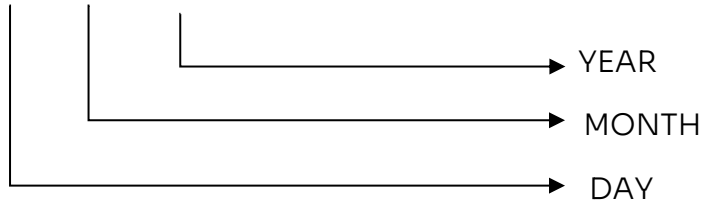
Table 10-1

Classification	Description
A	Content and drawing list
B	Control panel layout, location, sizes, etc
C	Power Circuit and DC control circuit
*D	PC interface, PLC interface, other interface, etc
*E	PLC Special I/O, Digital I/O, Analog I/O, etc
F	Motor Control Circuit for Stepper, Servo & Driver
*G	Motor Control Circuit like conveyor, speed control, AC or DC
*H	Control, connection for solenoid valve
I	Input, output terminal connection for sensor, SV, etc
*J	Module drawing consists of motors, sensors, solenoid valve, etc

*Not commonly used

10.1.3 Schematics Date Format

15 - 12 - 2019



TITLE : TAPC 1&2 CONNECTION				
DRAWING NO : B3065-CC012			REV: 0	DATE: 15/12/2019
	INITIAL	SIGN	DATE	 THIS IS A PROPERTY OF AEM SINGAPORE PTE LTD IT SHOULD NOT BE COPIED OR REPRODUCED WITHOUT THE WRITTEN APPROVAL OF AEM SINGAPORE PTE LTD.
DRAWN	PETHUEL ENOT		09/07/2019	
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APPROVED	ARUN KUMAR		26/10/2019	
SIZE : A3	DO NOT SCALE DRAWING	PAGE NO : 44 - 50		



10.2 Electrical / Schematic Circuits Content

This section consists of the electrical / schematic diagrams for the AEM Thermal Station.

Table 10-2

No	Drawing No	Title
1	B7144-CC001	BENCHTOP CONTROLLER BOX INCOMING POWER
2	B7144-CC002	BENCHTOP POWER SUPPLY 1,2 & 3 CONNECTIONS
3	B7144-CC003	BENCHTOP POWER SUPPLY 4 CONNECTION
4	B7144-CC004	BENCHTOP POWER SUPPLY 5 CONNECTION
5	ECA60387	CABLE ASSY APL DK CN1B VALVE 22 WAY
6	ECA60388	CABLE ASSY APL DK CN2B HEATER/RTD 24 WAY
7	B5461-CB001	THERMAL CONTROL BOARD LAYOUT
8	ECA60389	CABLE ASSY APL DK J1 HDM
9	ECA60390	CABLE ASSY APL DK RJ45 MICROCONTROLLER
10	ECA60391	CABLE ASSY APL DK P3 LCD DISPLAY
11	ECA60427	CABEL ASSY APL DK P4 FOR HEATER JACKET
12	ECA60392	CABLE ASSY APL DK P6 12VDC INCOMING PWR
13	ECA60393	CABLE ASSY APL DK P8 TELEMETRY
14	ECA60396	CABLE ASSY APL DK P11 LOW/HIGH VOLTAGE
15	ECA60397	CABLE ASSY APL DK P16 PRESSURE REGULATOR
16	ECA60398	CABLE ASSY APL DK P17 DIGITAL OUTPUT
17	ECA60428	CABLE ASSY APL DK TCO HDM-HV BD
18	ECA60401	CABLE ASSY APL DK P21 RS232



No	Drawing No	Title
19	ECA60403	CABLE ASSY APL DK RESET PUSH BUTTON
20	ECA60404	CABLE ASSY APL DK RESET BUTTON TO CTRL
21	EEA43183	ELECTRICAL ASSY APOLLO MS INDICATOR LED
22	ECA60525	CABLE ASSY APL DK FPGA P1 HDM CONTROL BD
23	ECA60526	CABLE ASSY APL DK POWER HDM-HV BD
24	EEA60429	ELECTRICAL ASSY APL DK FAN 1
25	ECA61282	CABLE ASSY BENCHTOP CN1 RCPT 22 WAY
26	ECA61301	CABLE ASSY BENCHTOP PUSH BUTTON
27	ECA61302	CABLE ASSY BENCHTOP RET/EXT SENSOR
28	ECA61283	CABLE ASSY BENCHTOP CN2 RCPT 24 WAY
29	ECA60402	CABLE ASSY RS232 CONNECTIONS
30	ECA60809	CABLE ASSY APL TELEMETRY CONNECTIONS
31	ECA60810	CABLE ASSY APL RJ45 CONNECTIONS
32	ESA61437	SENSOR ASSY BENCHTOP RETRACT SENSOR
33	ESA61438	SENSOR ASSY BENCHTOP EXTEND SENSOR
34	ECA61439	CABLE ASSY BENCHTOP RETRACT VALVE
35	ECA61440	CABLE ASSY BENCHTOP EXTEND VALVE
36	ESA61570	SENSOR ASSY BENCHTOP PROPORTIONAL VALVE
37	ESA61441	SENSOR ASSY BENCHTOP LOAD CELL SENSOR
38	ECA61110	CABLE ASSY BENCHTOP POWER HEATLET RCPT
39	ECA61568	CABLE ASSY BENCHTOP EEPROM/RTD RCPT
40	ECA61111	CABLE ASSY BENCHTOP TCO RCPT



No	Drawing No	Title
41	ECA61112	CABLE ASSY BENCHTOP HEATER JACKET RCPT
42	EWA61280	WIRE ASSY BENCHTOP IEC 14(L) TO SW1(1A)
43	EWA61279	WIRE ASSY BENCHTOP IEC 14(N) TO SW1(4B)
44	EWA61281	WIRE ASSY BENCHTOP IEC 14(PE) TO TB-PE
45	EWA60995	WIRE ASSY GND TB-FG TO MASTER BOARD
46	EWA60992	WIRE ASSY APL DK SW1(2A) L1-1
47	EWA60993	WIRE ASSY APL DK SW1(5B) N-1
48	EWA60994	WIRE ASSY APL DK R1 L-11
49	EWA60999	WIRE ASSY CONNECTOR IEC 14(L) L-3
50	EWA61000	WIRE ASSY APL DK 12V PWR SUPPLY
51	EWA61001	WIRE ASSY CONNECTOR IEC 14(N) TO TB2
52	EWA61297	WIRE ASSY APL DK PS5 TO TB -FG
53	EWA61567	WIRE ASSY GROUND TB-PE
54	ECA61104	CABLE ASSY BENCHTOP TTV KIT HEATLET
55	ECA61105	CABLE ASSY BENCHTOP TTV KIT RTD
56	ECA61106	CABLE ASSY BENCHTOP TH POWER HEATLET
57	ECA61107	CABLE ASSY BENCHTOP TH TCO
58	ECA61569	CABEL ASSY BENCHTOP EEPROM/RTD PLUG



NOTES

[illegible]



11. Maintenance



11.1 AEM Thermal Station Preventive Maintenance

Two Persons Required When Commencing Maintenance and Servicing

When doing maintenance and / or repairing work, two people are required to be present for precautionary purpose. This shall be done so that one of them can aid in the lookout for the push button panel, preventing other people from accidentally pushing the button while the other person is servicing the machine.

All employees, upon observing a machine or piece of equipment which is locked out to perform servicing, maintenance, or replacement of parts, shall not attempt to start, energize, or use that machine or equipment.

Before doing any maintenance or replacement of parts, power off the Thermal Station (refer to [6.2.3 Power OFF Procedure](#) and [2.5.6 Standard 3 Point Safety Check](#)). Follow all the safety procedures before start doing any maintenance and replacement (refer [2.2.2 General Safety Guidelines](#)) and wear all PPE (refer [2.2.3 Personal Safety Equipment](#)).

Spring Washers

Any time a Spring Washer is removed in the performance of regular maintenance activities, replace the Spring Washer with a new washer. Spring Washers should not be reused. Record any replaced Spring Washers in available procedure checklist.

General Maintenance

Specific machine components should be checked periodically to ensure they are working properly. Follow the preventive maintenance (PM) procedures given below will facilitate in minimizing production interruptions.

The procedures for regular inspections and maintenance can be classified into intervals of:

1. Semi-Annually (every 6 months)



ONLY QUALIFIED PERSONNEL SHOULD CONDUCT MAINTENANCE OPERATIONS.
ALWAYS DISCONNECT THE MACHINE FROM MAINS AND COMPRESSED AIR
BEFORE DOING MAINTENANCE OR CLEANING OPERATION.



ENSURE THAT A SIGN IS PLACED INDICATING THAT THE EQUIPMENT AND ROBOT IS
BEING SERVICED OR ADJUSTED, IN ORDER TO KEEP ANYONE FROM TOUCHING THE
OPERATIONAL PANEL.



ALWAYS WEAR SAFETY GLASSES WHENEVER WORKING WITH COMPRESSED AIR. DIRECT AIR PRESSURE AND/OR FLYING DEBRIS FROM AIR BLOWN PARTICLES COULD CAUSE SERIOUS EYE INJURIES.

11.1.1 Semi Annual Preventive Maintenance

Semi Annual Preventive Maintenance to be performed every 6 months.

11.1.1.1 Warning Label Check

Refer to [2.2 Safety Alert Boxes](#). Check all warning label condition and adhesion. Replace any label that is not legible or is peeling away from panel surface.

11.1.1.2 Tamper Proof Screws

Check all Tamper Proof Screws and verify Tamper Proof Screws are in place and torqued correctly. Replace any screws that should be tamper proof but has been replaced with a standard screw or if the Tamper Proof Screw has been damaged. Torque to the specification by screw size.

11.1.1.3 Spring Washers

Any time spring washer is removed in the performance regular maintenance activities replace the spring washer with a new washer. Spring washers should not be rushed. Record any replaced spring washers in available procedure checklist.



NOTES

[illegible]



12. Appendix

**12.1 Material Safety Data Sheet****12.1.1 Isopropyl Alcohol****MATERIAL SAFETY DATA SHEET
ISOPROPYL ALCOHOL**

Ref: AK/002/IPA

CHEMICAL NAME

2- Propanol.

TYPICAL PROPERTIES

Density @ 20 °C		0.785 – 0.786 kg/l
Flash point, Abel		10°C
Auto-ignition temperature		425°C
Explosive Limits in Air,	Lower	2 % vol.
	Upper	12 % vol.
Vapour Density @ 0°C & 1 atm.		2.1
Threshold Limit Value,	TLV-TWA	400 ppm (985 mg/m ³)
ACGIH 1989-90)	TLV-STEL	500 ppm (1230 mg/m ³)

HAZARDOUS PROPERTIES

Isopropyl alcohol (IPA) is a highly flammable liquid and can form explosive mixture with air particularly in emptied uncleaned receptacles. The vapour is invisible, heavier than air and spreads along the ground; distant ignition is possible. Do not use compressed air for filling, discharging, or handling. IPA is irritating to the eyes and mucous membranes in liquid and vapour form. Generalized toxicity is similar to ethyl alcohol e.g. inebriation. Prolonged or repeated skin contact may cause dermatitis.



GENERAL PRECAUTIONS

Isolate from sources of heat, naked flames and sparks. Maintain adequate ventilation and avoid breathing the vapour. Avoid contact with the skin, eyes and clothing.

Wearing protective equipment as and when appropriate.

PROTECTIVE CLOTHING AND EQUIPMENT

Normal Operations.

Eye : Chemical goggles or full-face shield with forehead protection.

Hand : Wrist length gloves, nitrile rubber, PVC, neoprene.

Foot : Safety shoes or boots: chemical resistant.

Body : Standard issue work clothes.

Respiratory protection: Should be handled in a well-ventilated area.
Wear a full-face respirator with organic vapour cartridge if there is an inhalation risk. (Nominal Protection Factor NPF 20).

Spillage

Eye : Covered by respiratory protection.

Hand : Wrist length gloves: Nitrile rubber, PVC, neoprene.

Foot : Safety boots: knee length.

Body : Chemical resistant PVC one-piece suit.

Respiratory protection: Full-face piece respirator with organic vapour cartridge (Nominal protection factor NPF 20). If spillage occurs in a poorly ventilated area, wear compressed air line or self-contained breathing apparatus. (NPF 2000)



ACTION IN CASE OF SPILLAGE / LEAKAGE

Extinguish all naked lights and remove ignition sources.

Avoid sparks. Evacuate personnel from the area. Ventilate the contaminated area thoroughly.

Avoid inhaling the vapour. Wear appropriate protective clothing.

Small Spillage

Absorb the liquid with sand or earth.

Shovel up and place in a labelled sealable container for subsequent disposal.

Flush contaminated area with water.

Large Spillage

Prevent spillage from spreading by use of sand or earth.

Transfer spillage to a salvage tank if possible; otherwise treat as for a small spillage.

Inform local authorities (eg. fire service) immediately if the spilt liquid enters the surface water drains since a potential explosive and toxic hazard will exist.

Leakage

Shut off leaks if without risk.

Put leaking drums in safe containers (overdrum). Otherwise treat as for appropriate spillage.



NOTES

[illegible]